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Multi-armed Bandit Experimental Design: Online Decision-Making and Adaptive Inference

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Abstract. Multi-armed bandit has been well known for its efficiency in online decision-making in terms of minimizing the loss of the participants' welfare during experiments (i.e., the regret). In clinical trials and many other scenarios, the statistical power of inferring the treatment effects (i.e., the gaps between the mean outcomes of different arms) is also crucial. Nevertheless, minimizing the regret entails harming the statistical power of estimating the treatment effect because the observations from some arms can be limited. In this paper, we investigate the trade-off between efficiency and statistical power by casting the multi-armed bandit experimental design into a *minimax multi-objective optimization problem*. We introduce the concept of *Pareto optimality* to mathematically characterize the situation in which neither the statistical power nor the efficiency can be improved without degrading the other. We derive a useful *sufficient and necessary condition* for the Pareto optimal solutions to the minimax multi-objective optimization problem. Additionally, we design an effective Pareto optimal multi-armed bandit experiment that can be tailored to different levels of the trade-off between the two objectives. Moreover, we extend the design and analysis to the setting where the outcome of each arm consists of an adversarial baseline reward and a stochastic treatment effect, demonstrating the robustness of our design. Finally, motivated by clinical trials, we examine the setting where the employed experiment must split the experimental units into a small number of batches, and we propose a flexible Pareto optimal design.

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Keywords: adaptive experimental design • multi-armed bandit • minimax multi-objective optimization • Pareto optimality

1. Introduction

Multi-armed bandit (MAB), one of the most effective frameworks for sequential decision making, is renowned for its adaptability as more information becomes available. The adaptive allocation of experimental units has been demonstrated to be more efficient than some traditional random experiments, such as classical randomized, controlled trials (RCTs) in Lai and Robbins (1985), and has been extensively used in a variety of applications. The current theoretical literature has studied MAB extensively, mainly focusing on understanding the best achievable efficiency in the hope of minimizing the loss of the participants' welfare during experiments (i.e., the regret). The optimal regret of stochastic MAB has been well understood to be $\Theta(\log n)$ (see, e.g., Lattimore and Szepesvári 2020), which can be achieved by a series of well-known algorithms, including the upper confidence bound (UCB)-based algorithms and Thompson sampling (TS)-based algorithms.

However, for many real-world problems, regret is not the only metric that matters when conducting experiments. Consider the following hypothetical scenario as a motivating example where a drug company is using the MAB to evaluate the efficacy of a new drug. Regret measures the overall loss of patients' welfare during the experiment, so we want to keep it to a minimum. Especially for rare or fatal diseases, it is expected to treat the patients within the trial as effectively as possible. Additionally, it is always crucial to report what kind of difference the new treatment can make compared with the control (i.e., to infer the average treatment effect (ATE)) (see, e.g., Imbens and Angrist 1994, Heckman and Vytlacil 2001, Aronow and Samii 2017). It is also of great interest to understand the difference between any two drugs

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if more than one is being tested simultaneously. This kind of inference can be quite instructive in that it may be used to determine which alternative should be used when the best drug is unavailable because of shortage, regulations, or other factors. In another example from political science, Caria et al. (2024) design a field experiment to test policies to help Syrian refugees and local jobseekers in Jordan to find work. They discuss the importance of balancing the goals of maximizing the precision of treatment effect estimates and maximizing the welfare of experimental participants. Another example from online education discussed in Zhang et al. (2020) is that the practitioners want to gain generalizable knowledge about the different performances of new teaching strategies, not just determine which one is the best. These three examples illustrate *the necessity of adaptive statistical inference of ATE while making online assignment decisions* in MAB experiments.

Online decision making and statistical inference of ATE have been extensively investigated separately, but when they are jointly considered, several new and crucial challenges arise. First, MAB algorithms collect data adaptively, and consequently, it is not appropriate to consider the collected data as independent and identically distributed (i.i.d.). This imposes extra challenges when conducting inference, and it may harm the statistical power. For example, using the sample average to estimate ATE (as practitioners usually do in RCTs) in some MAB algorithms, including UCB and TS, creates a nonnegligible bias, as pointed out by much work in the literature (see Xu et al. 2013, Luedtke and Van Der Laan 2016, Nie et al. 2018, Hadad et al. 2021). In other words, MAB algorithms are effective in learning the optimal online decision-making policy but lack statistical power. In contrast, some existing well-established ATE estimation methods are recognized for their strong statistical power, but they will incur significant regrets. Specifically, because a predetermined percentage of the population will be assigned to the suboptimal arm, the regret of a traditional RCT will increase linearly with the number of samples. Another commonly adopted assumption for adaptive ATE estimators is "overlap," which states that the probability of playing all arms is lower bounded away from zero by a universal constant over time (see, e.g., Hahn et al. 2011, Khan and Ugander 2021). Then, the accumulative regret is also linear because of the linear growth of the expected number of times the suboptimal arms are played. One of our goals in this paper is *to design MAB experiments with both online decision-making policies and adaptive inference for ATE with finite sample guarantees* by harnessing the efficiency and statistical power from each side.

Moreover, the relationship between the two tasks, adaptive inference and online decision making, may change throughout the experiment. On the one hand, these two tasks can complement one another, particularly in the initial phase of the experiment. An accurate estimation of ATE is undoubtedly informative for learning the best decision-making policy. An efficient online decision-making algorithm typically explores all arms actively, including the suboptimal ones, at the beginning explicitly or implicitly in order to gather enough information. This is advantageous for inference as well. On the other hand, the success of gauging ATE may be hindered by online decision making, which is only concerned about being able to identify the best arm without considering how much better it is. To be more precise, minimizing the regret requires the algorithm to stop performing the suboptimal action as soon as it gains enough confidence about the suboptimality. Because of the potentially relatively few observations, it will make an accurate estimation of the suboptimal arm virtually impossible. However, because ATE is the difference between two arms, an accurate inference heavily relies on the estimators of both arms, and thus, the statistical power of estimating ATE may be limited by efficient online decision making. Therefore, there exists an important trade-off between these two tasks. In order to describe such a trade-off, we introduce the term "Pareto optimality" to characterize the circumstance where neither regret nor estimating error of ATE can be made better off without making the other worse off. *How to statistically quantify and practically achieve the Pareto optimality in MAB experiments* remains an open question.

1.1. Preliminaries

We start by establishing some definitions and our problem settings. In MAB experiments, there is a finite set $\mathcal{A}$ of arms (i.e., treatments or actions) $a \in \mathcal{A}$ with $|\mathcal{A}| = K$. Without loss of generality, the control can also be seen as an action. n is the total number of experimental units (or the time horizon). At each time $t \le n$, the environment generates a reward $r_t(a)$ for every arm $a \in \mathcal{A}$. After choosing arm a_t, only the reward of the chosen arm $r_t := r_t(a_t)$ can be observed. In stochastic MAB experiments, the expectation $\mathbb{E}[r_t(a)] = \mu_a \in [-1, 1]$, where μ_a is the unknown true reward of arm a, which is disturbed by an i.i.d. noise to generate $r_t(a) \in [-1, 1]$. A stochastic MAB instance can be denoted by $\nu = (P_1, \ldots, P_k)$, where P_i is the distribution of the rewards of arm i. The optimal arm is the arm with the maximum true reward, which is denoted by $a^* := \arg\max_{a \in \mathcal{A}} \mu_a$, and thus, it is also unknown. We define the gap between arm i and arm j as $\Delta^{(i,j)} := \mu_i - \mu_j$, for any $i \ne j \in [K]$. Among all $\Delta^{(i,j)}$, we additionally define the difference between μ_{a^*} and μ_a as the suboptimality gap (i.e., $\Delta(a) := \mu_{a^*} - \mu_a$ for $a \in \mathcal{A} \backslash \{a^*\}$). Specifically, when $K = 2$, we can define arm 1 to be the treatment of interest and arm 2 to be a control. If the treatment has a positive average effect among the population (i.e., $\mu_1 > \mu_2$), $\Delta^{(1,2)} = \Delta(2)$ is the ATE. Otherwise, $\Delta^{(1,2)} = -\Delta(1)$ is the ATE that we want to estimate. Therefore,

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from now on, we only need to focus on $\Delta^{(i,j)}$ as the ATE follows naturally by the definition of each arm when designing the experiments. In this paper, we will elaborate on $|\Delta^{(i,j)}| = \Theta(1)$ for all $i \neq j \in [K]$, which is arguably the most fundamental case in real applications. We will relax such an assumption in Section 7. Denote all stochastic MAB instances satisfying the mentioned assumptions to constitute a feasible set $\mathcal{E}_0$.

At every time t, the decision maker observes the history $\mathcal{H}_t = (a_1, r_1, \ldots, a_t, r_t)$. An admissible policy $\pi = \{\pi_t\}_{t\geq 1}$ maps the history $\mathcal{H}_{t-1}$ to an action a_t. Denote the probability with which arm a is chosen at time t as $\pi_t(a) = \mathbb{P}(a_t = a|\mathcal{H}_{t-1})$ under policy π. To measure the efficiency of online learning, we use the accumulative *regret*, which is defined as the expected difference between the reward under the optimal policy and the policy π (i.e., $\mathcal{R}(n,\pi) = \mathbb{E}^{\pi}[nr_t(a_t^*) - \sum_{i=1}^{n} r_i(a_i)]$). In stochastic MABs, $a_t^* = a^*$ for all t. In addition, an admissible adaptive estimator $\hat{\Delta}^{(i,j)} = \{\hat{\Delta}_t^{(i,j)}\}_{t\geq 1}$ maps the history $\mathcal{H}_t$ to an estimation of $\Delta^{(i,j)}$ at each time t. We use the *error* defined as the expected distance of $\Delta^{(i,j)}$ and $\hat{\Delta}_t^{(i,j)}$ (i.e., $e(t, \hat{\Delta}^{(i,j)}) = \mathbb{E}[|\Delta^{(i,j)} - \hat{\Delta}_t^{(i,j)}|]$) to measure the quality of the estimation. In the causal inference literature, the length of a valid confidence interval is frequently utilized as a metric to assess statistical power. In the main text, our focus will initially be on the estimation error and illustrating the process of constructing confidence intervals. Furthermore, we will explore the extension of our findings when prioritizing the length of the confidence interval as our primary objective rather than the error. We define $\hat{\Delta} := \{\hat{\Delta}(i,j)\}_{i<j\leq K}$ to represent all of the estimators on the gap between any two arms. A design of an MAB experiment can then be represented by an admissible pair $(\pi, \hat{\Delta})$. Different from traditional MAB problems, the optimal design of MAB experiments in this paper is solving the following minimax multi-objective optimization problem:

$$\min_{(\pi,\hat{\Delta})} \max_{\nu\in\mathcal{E}_0} \left(\mathcal{R}_\nu(n,\pi),\ \max_{i<j\leq K} e_\nu(n, \hat{\Delta}^{(i,j)})\right), \tag{1}$$

where we use the subscript ν to denote the MAB instance. Equation (1) mathematically describes the two goals: minimizing the regret and the inference error under the worst case. For traditional MAB problems, $\min_\pi \max_{\nu\in\mathcal{E}_0} \mathcal{R}(n,\pi)$ is the main objective. The asymptotic behavior of $\hat{\Delta}$ is one of the central focuses of the existing ATE literature. Note that π and $\hat{\Delta}$ are complicatedly correlated through the history $\mathcal{H}$, and thus, the optimization problem in Equation (1) has some implicit constraints on π and $\hat{\Delta}$. Such kinds of constraints may lead to a complicated feasible region and thus, impose great challenges on solving the optimization problem.

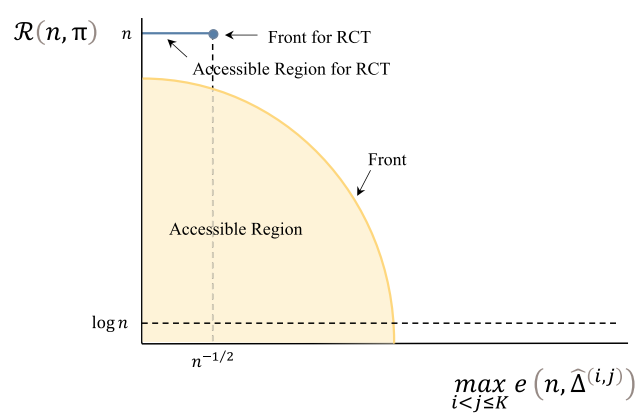
In order to define the optimality and understand the structure of solutions, we start from the inner maximization over all $\nu \in \mathcal{E}_0$ problems in Equation (1). Intuitively, under a given $(\pi, \hat{\Delta})$, the pairs $(\mathcal{R}_\nu, \max_{i<j\leq K} e_\nu)$ for all $\nu \in \mathcal{E}_0$ can constitute an accessible region, and the *front* of the accessible region can be defined to be the optimal values of the inner maximum problem (see Figure 1). Formally, we define the front of a given pair $(\pi, \hat{\Delta})$ as Definition 1.

Definition 1 (Front). The front for a given pair $(\pi, \hat{\Delta})$, denoted by $\mathcal{F}(\pi, \hat{\Delta})$, is a set of pairs of $(R, e) \in \mathbb{R}^+ \times \mathbb{R}^+$ satisfying

1. $\exists \nu_0 \in \mathcal{E}_0$, $(\mathcal{R}_{\nu_0}(n,\pi), \max_{i<j\leq K} e_{\nu_0}(n, \hat{\Delta}^{(i,j)})) = (R, e)$;
2. $\nexists \nu \in \mathcal{E}_0$, $\mathcal{R}_\nu(n,\pi) \geq R$ and $\max_{i<j\leq K} e_\nu(n, \hat{\Delta}^{(i,j)}) > e$; and
3. $\nexists \nu \in \mathcal{E}_0$, $\mathcal{R}_\nu(n,\pi) > R$ and $\max_{i<j\leq K} e_\nu(n, \hat{\Delta}^{(i,j)}) \geq e$.

In this definition and Definition 2, we abuse the symbols $=$, $\geq$, and $>$ for brevity. When we compare two values here, we only focus on their dependence on n.[1] The first condition ensures the achievability of the front. The second and third conditions are describing that there does not exist any instance that will incur no fewer values on both objectives and a strictly larger value on at least one. From now on, including in Definition 1, when it comes to comparing regrets or errors, we only focus on the order of n ignoring the universal constant and the logarithm terms because n is usually relatively large. In Figure 1, the yellow region and its boundary are an example of the accessible region and the front, respectively. We also present the traditional RCTs in Figure 1. Because $|\Delta^{(i,j)}| = \Theta(1)$ for the instance class $\mathcal{E}_0$, the RCTs will usually incur linear regrets, and thus, the accessible region is a line. By the theoretical results for RCTs (see, e.g., Wainwright 2019), the best achievable accuracy is $\Theta(n^{-1/2})$, and thus, the front for RCTs is at the point $(n, n^{-1/2})$, ignoring the constant independent of n and the logarithm terms. Additionally, the well-known results in MAB tell that the worst-case regret of any policy is no smaller than $\log(n)$, and thus, all accessible regions will inevitably have some parts on or above the $\log(n)$ line shown in Figure 1.

Figure 1. (Color online) Examples of Accessible Regions and Fronts



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In order to define the optimality for our minimax objective (1), we first define the *Pareto dominance* between two feasible solutions based on the definition of the front as follows.

Definition 2 (Pareto Dominance). A feasible solution $(\pi_1, \hat{\Delta}_1)$ is said to *Pareto dominate* another solution $(\pi_2, \hat{\Delta}_2)$ if $\forall (R_1, e_1) \in \mathcal{F}(\pi_1, \hat{\Delta}_1)$, $\exists (R_2, e_2) \in \mathcal{F}(\pi_2, \hat{\Delta}_2)$ such that at least one of the following two conditions holds: (i) $R_1 \le R_2$ and $e_1 < e_2$ or (ii) $R_1 < R_2$ and $e_1 \le e_2$.

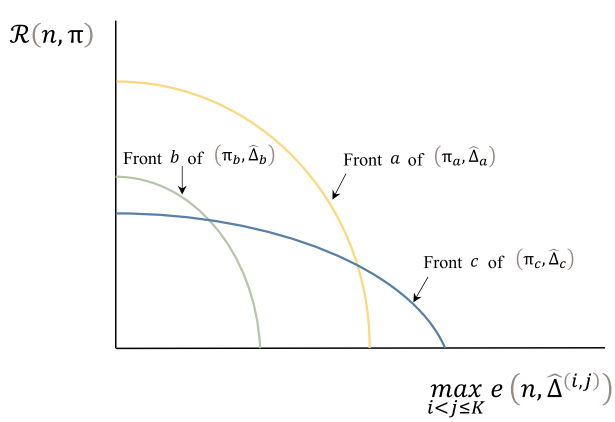
The definition formally describes that $(\pi_1, \hat{\Delta}_1)$ is Pareto better than $(\pi_2, \hat{\Delta}_2)$ if for any point (R_1, e_1) on the front of $(\pi_1, \hat{\Delta}_1)$, there exists some point (R_2, e_2) on the front of $(\pi_2, \hat{\Delta}_2)$ such that (R_1, e_1) is no larger than (R_2, e_2) on both coordinates and is strictly better on at least one coordinate. In Figure 2, we present toy examples of the fronts of three different solutions. By the definition, $(\pi_b, \hat{\Delta}_b)$ Pareto dominates $(\pi_a, \hat{\Delta}_a)$, and $(\pi_c, \hat{\Delta}_c)$ can neither Pareto dominate nor be Pareto dominated by $(\pi_a, \hat{\Delta}_a)$ or $(\pi_b, \hat{\Delta}_b)$. Based on such Pareto dominance, we can have the following definition of the crucial concept in the paper *Pareto optimality*.

Definition 3 (Pareto Optimality). An admissible pair of $(\pi^*, \hat{\Delta}^*)$ is *Pareto optimal* in terms of the dependence of n if it is not Pareto dominated by any other solution. *Pareto frontier* denoted as $\mathcal{P}$ is the envelop of the fronts of all of the Pareto optimal solutions.

Here, because we focus on the order of n when we compare the values in Definitions 1 and 2, our Pareto optimality also follows such a philosophy and is defined to be in terms of dependence of n. Such a simplification is a common and standard practice in the MAB literature, and we discuss the influence of ignoring the constant in Section 7. That an admissible pair $(\pi^*, \hat{\Delta}^*)$ is Pareto optimal does not mean that it can Pareto dominate every other policy, and thus, there may exist a group of admissible pairs that are all Pareto optimal.

In general, designing Pareto optimal MAB experiments is at the center of this work. The first natural research question is *how to solve the minimax multi-objective optimization problem to get the Pareto optimal solutions.* The first challenge stems from the multiple objectives, which are arguably more challenging to tackle than solving the single-objective optimization problem, like the traditional MAB problems. Moreover, π and $\hat{\Delta}$ have the measurability constraints and are highly correlated through the history $\mathcal{H}_t$. These constraints are hard to explicitly integrate into the optimization problem and endow the feasible region with complicated structures. This hinders the employment of the existing methods in the minimax multi-objective optimization literature. Furthermore, finding only one Pareto optimal solution is always not enough. It is important to design experiments flexibly under different requirements for the trade-off between these two objectives. This is indeed asking *how to obtain the optimal Pareto optimal solutions given different levels of trade-off in these two objectives*.

Figure 2. (Color online) Examples of Pareto Dominance



1.2. Contributions and Main Results

The main contribution of this paper is the Pareto optimal design of MAB experiments, especially the statistical understanding of the trade-off between online decision making and adaptive inference in MAB experiments. To the best of our knowledge, this work is the first to jointly consider the efficiency and statistical power in the experimental design literature. We are also the first to introduce the minimax multi-objective optimization framework to experimental designs, which also greatly generalizes the existing minimax optimization framework in the MAB literature. We next summarize our main results and highlight our contributions from four folds.

First, we find a *sufficient and necessary condition* for the Pareto optimal solutions of the minimax multi-objective optimization Problem (1). Specifically, an admissible pair $(\pi^*, \hat{\Delta}^*)$ is Pareto optimal if and only if

$$\max_{\nu \in \mathcal{E}_0} \left[\left(\max_{i<j\le K} e_\nu(n, \hat{\Delta}^{*(i,j)}) \right) \sqrt{\mathcal{R}_\nu(n, \pi^*)} \right] = \tilde{\mathcal{O}}(1), \qquad (2)$$

which means that for any Pareto optimal solution, the product of the error and the square root of the regret in the worst case is no larger than a constant order independent of n. Compared with the definition of Pareto optimality, this condition is much simpler to interpret and is likely to be easier to verify. In order to show such equivalence, technically, we establish an information-theoretical minimax lower bound to portray the trade-off between these two objectives. Specifically,

$$\inf_{(\pi, \hat{\Delta})} \max_{\nu \in \mathcal{E}_0} \left[\left(\max_{i<j\le K} e_\nu(n, \hat{\Delta}^{(i,j)}) \right) \sqrt{\mathcal{R}_\nu(n, \pi)} \right] = \Omega(1). \qquad (3)$$

This lower bound tells us that no solution can do better on $(\max_{i<j\le K} e_\nu(n, \hat{\Delta}^{(i,j)}))\sqrt{\mathcal{R}_\nu(n, \pi)}$ than a constant order in the worst case. Particularly, because UCB/TS

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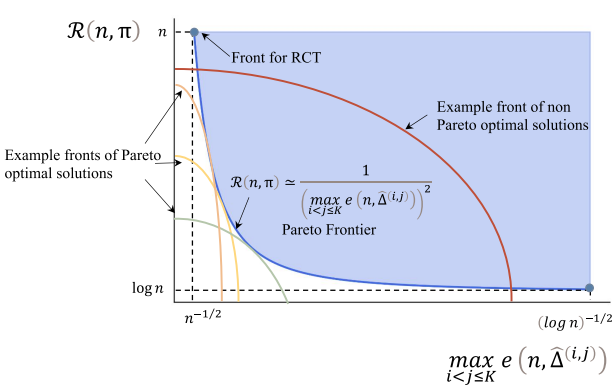
algorithms have the regret upper bound of $\mathcal{O}(\log n)$, then any ATE estimator based on the UCB/TS algorithm cannot avoid an error of $\Omega(1/\sqrt{\log n})$ in the worst case. This explicitly shows the lack of statistical power of UCB/TS algorithms for ATE inference. In Figure 3, we give some examples of the fronts of Pareto optimal and suboptimal solutions. By Equation (3), the front of any admissible policy has intersections with the region $(\max_{i<j\le K} e(n, \hat{\Delta}^{(i,j)}))\sqrt{\mathcal{R}(n,\pi)} \ge 1$ (the blue region in Figure 3). Equation (2) indicates that the fronts of the Pareto optimal solutions only have the intersections on the boundary of the blue region in Figure 3. The nonoptimal solutions have some hard instances falling into the interior of the blue region in Figure 3.

Second, we propose an efficient Pareto optimal design for stochastic MAB experiments that can adapt to different levels of trade-off between the two objectives. Specifically, we combine the well-known EXP3 algorithm (see, e.g., Auer et al. 2002a, Seldin et al. 2013, Lattimore and Szepesvári 2020) and the idea of extra forced exploration, which means that the algorithm purposely plays the other arms after it identifies the best one. For any given $\alpha \in [0,1]$ as input, we prove that the regret of our algorithm is $\tilde{\mathcal{O}}(n^{1-\alpha})$ and that the estimation error of ATE is $\mathcal{O}(1/\sqrt{t^{1-\alpha}})$ for all $t \le n$, which satisfies Condition (2) by taking $t = n$. Note that α balances the two objectives. If α is large, the practitioners emphasize the control of regret. Small α will lead to a more accurate ATE estimation. Furthermore, we demonstrate the construction of an anytime confidence sequence for our estimator and present a valid approach to hypothesis testing as the traditional t test may not be suitable in our particular setting.

Third, we extend the design and analysis of stochastic MAB experiments to a setting where the outcome consists of an adversarial baseline reward and a stochastic treatment effect. In many applications, the i.i.d.-generated reward assumption may be too restrictive. On the other hand, the adversarial MABs are at the other extreme end because a commonly held conviction is that treatment effects may be more structured than the outcomes themselves (see Kennedy 2020, Carranza et al. 2023). To bridge the gap between the stochastic and adversarial MABs in modeling, we examine a more general reward structure, where the rewards can be decomposed into an adversarial baseline part and a stochastic treatment effect part. We find that our design developed for stochastic MAB experiments can be directly deployed to this case without any modification and that it has the same performance guarantee. Furthermore, by some mild adjustments of the formulation, the fully adversarial setting can also be covered, demonstrating the robustness of our design.

Finally, in light of practical application, especially clinical trials, we investigate the experiments where the employed policy must split trails into a limited number of batches as seen in previous studies, such as Perchet et al. (2016) and Gao et al. (2019). Additionally, certain outcomes, such as five-year survival rates or vaccine effectiveness, cannot be observed immediately after the treatment and are revealed to the decision maker in a batch form. To address this case, we propose a Pareto optimal design for batched experimental design. Our results show that with a limited number of batches, the batched MAB experiments can provide performances competitive with the fully stochastic MAB experiments. Moreover, we demonstrate that the batched MAB experiments can serve as an bridge between RCTs and the stochastic MAB experiments.

Figure 3. (Color online) Summary of the Results for Stochastic MAB Experiments



1.3. Related Work

1.3.1. Learning Efficiency in MABs. A main body of literature in MABs has focused on the online learning efficiency (i.e., minimizing regret). Two representative classes of algorithms that can provide optimal regret bounds (i.e., $\Theta(\log n)$ regret bounds) are UCB-based algorithms (see, e.g., Lai et al. 1985, Agrawal 1995, Auer 2002, Auer et al. 2002a, Carpentier et al. 2011, Garivier and Cappé 2011) and TS-based algorithms (see, e.g., Thompson 1933, Chapelle and Li 2011, Kaufmann et al. 2012, Russo and Van Roy 2016). Fan and Glynn (2021) and Simchi-Levi et al. (2022) reveal that efficiency-optimized bandit algorithms may suffer from serious heavy-tailed risk. Additionally, the learning efficiency of batched MAB also received a lot of academic attention (see, e.g., Perchet et al. 2016, Gao et al. 2019, Esfandiari et al. 2021, Jin et al. 2021, Agarwal et al. 2022). The batched MAB is also closely related with the bandit problems with switching constraints (see, e.g., Agrawal et al. 1988, Cesa-Bianchi et al. 2013, Simchi-Levi and Xu 2023). In this paper, our design is based on the idea of EXP3, which was initially designed for adversarial MABs by Auer et al. (2002b). Recently, Seldin et al. (2012, 2013) improved the analysis of the EXP3 algorithm and achieved the logarithmic optimal regret in

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stochastic environments. Another growing body of the MAB literature is aiming at identifying the best arm (see, e.g., Jennison et al. 1982, Mannor and Tsitsiklis 2004, Chan and Lai 2006, Gabillon et al. 2012, Garivier and Kaufmann 2016). Zhong et al. (2021) study the trade-off between regret minimization and best-arm identification, and also, they adopt the concept of Pareto optimality.

1.3.2. Adaptive Experimental Design. Experimental design is becoming more and more popular in operations research, econometrics, and statistics (see, e.g., Johari et al. 2015, 2022; Aronow and Samii 2017; Eckles et al. 2017; Athey et al. 2018; Bojinov et al. 2021, 2023; Wager and Xu 2021b; Farias et al. 2022a, b; Xiong et al. 2024).

Adaptive experimental design is the area of study that is most pertinent to this work. MAB itself can also be seen as a type of adaptive experimental design. Scott (2015) summarizes the application of MAB experiment in the online service economy. Offer-Westort et al. (2021) study adaptive experiments to identify the best arm and estimate the mean outcome of the best arm. Kasy and Sautmann (2021) design experiments to identify the best arm when the data arrive in a batch fashion. Wager and Xu (2021a) provide a diffusion-asymptotic perspective to understand sequentially randomized experiments. Qin and Russo (2022) investigate bandit experiments where a potentially nonstationary sequence of contexts influences arms' performance.

Hahn et al. (2011) design a two-stage experiment for estimating ATE. To achieve the desired asymptotic normality, their design of the length of the two stages will incur undesirable linear regret in our context. Kato et al. (2020) investigate adaptive experiments for ATE when context can be observed. Atan et al. (2019) reveal the suboptimality of RCTs when the patients are recruited in cohorts rather than all at once. Glynn et al. (2020) propose a theoretical model to study optimal experimental design when temporal interference exists by transforming it into to a Markov decision problem. Bhat et al. (2020) consider the problem of optimally allocating test subjects to maximize the precision of estimating ATE in A/B test by solving dynamic programming.

1.3.3. Inference in MABs. The work from Villar et al. (2015) is a pioneering work in revealing that MAB algorithms offer significant advantages in assigning more patients to better treatments and severe limitations on resulting statistical power from an empirical perspective. In this work, we introduce the concept "Pareto optimal" to statistically describe and quantify such a trade-off.

There is a substantial literature on postexperiment inference from logged adaptively collected data where bandit-generated data are a typical example. One of the central tasks along this line is the evaluation of a new policy given historic data (see, e.g., Dudík et al. 2014, Li et al. 2015, Swaminathan and Joachims 2015, Wang et al. 2017, Kallus and Zhou 2018, Athey and Wager 2021, Zhan et al. 2021, Zhou et al. 2023). Chen et al. (2022) provide a procedure to adopt bootstrap on the data collected by a bandit to debias the sample means. A more relevant line to this work is adaptive inference in MABs. Hadad et al. (2021) construct confidence intervals for policy evaluation, including ATE, in adaptive experiments. As mentioned in their paper, there are some assumptions on the propensity score that may harm the learning efficiency. Dimakopoulou et al. (2021) focus on conducting inference on the true mean of each arm based on the data collected so far at each step with the optimal regret and outstanding empirical performances. Dimakopoulou et al. (2017, 2019) consider the case where context information exists and estimate the conditional expectation of each action's reward under different contexts. Han et al. (2022) recently consider the inference of a low-rank matrix under the setting of matrix contextual bandit.

1.3.4. Multitasking Bandits. This work is also closely related to an emerging field where the multitasking bandit is studied, where minimizing regret is not the only objective. Yang et al. (2017) regulate the false discovery rate while learning the best arm in MABs. Yao et al. (2021) want to ensure the capacity to detect whether an intervention has an effect in addition to using the contextual bandit to customize actions for consumers. The work of Erraqabi et al. (2017) is the most relevant to us because they also want to balance the trade-off between regret and estimation error. They redefine a new reward function based on the observed rewards and the error bounds, and they use such a new reward to guide online decision-making. In this way, they do not explicitly capture the trade-off between these two goals as we do and thus, cannot describe the optimality of their design.

Throughout this paper, we define $a \wedge b \triangleq \min\{a, b\}$ and $a \vee b \triangleq \max\{a, b\}$ for $a, b \in \mathbb{R}$, and we use $[n]$ to denote the set $\{1, 2, \ldots, n\}$ for any positive integer n. We use "arm," "action," and "treatment" interchangeably. We use "MAB" to denote the problem where regret is the only objective, and we use "MAB experiment" to denote the setting where online decision making and inference are jointly considered.

2. Stochastic MAB Experiments with $K = 2$

In this section, we consider the stochastic MAB experimental designs with two arms whose rewards are i.i.d. generated. We first establish a crucial lower bound and a sufficient condition for the Pareto optimality. Then, we propose a series of Pareto optimal designs satisfying the sufficient condition. The necessity of the condition is then demonstrated through the constructed designs.

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2.1. A Lower Bound and a Sufficient Condition

In this subsection, we are going to establish a lower bound for $(\max_{i<j\le K} e_\nu(n,\hat{\Delta}^{(i,j)}))\sqrt{\mathcal{R}_\nu(n,\pi)}$. For brevity, we ignore the superscripts of $\Delta^{(i,j)}$ and $\hat{\Delta}^{(i,j)}$ because there is no ambiguity when $K=2$. The following lemma presents a minimax lower bound for the error of any admissible estimator $\hat{\Delta}_n$ under a specified decision-making policy π.

Lemma 1. *When $K=2$, for any given online decision-making policy π, the error of any ATE estimators can be lower bounded as follows for any function $\phi: n \to [0,1/4]$ and any $u \in \mathcal{E}_0$:*

$$\inf_{\hat{\Delta}_n} \max_{\nu\in\mathcal{E}_0} \mathbb{P}_\nu(|\hat{\Delta}_n - \Delta_\nu| \ge \phi(n)) \ge \frac{1}{2}\left[1-\sqrt{\frac{16}{3}\phi(n)^2\frac{\mathcal{R}_u(n,\pi)}{|\Delta_u|}}\right]. \tag{4}$$

Lemma 1 demonstrates that for any $\hat{\Delta}_n$, there exists a hard instance $\nu\in\mathcal{E}_0$ such that the error is no smaller than $\phi(n)$ with some lower-bounded probability. For a fixed $\phi(n)$ and $u\in\mathcal{E}_0$, the right-hand side (RHS) of Equation (4) decreases with $\mathcal{R}_u(n,\pi)$. It is consistent with our observation that a larger regret leads to more observations from the suboptimal arm, which makes a better estimation of ATE possible. Additionally, $\phi(n)$ plays a delicate role in Equation (4) and needs to be chosen carefully. On the left-hand side, we expect $\phi(n)$ to be as large as possible. However, a large $\phi(n)$ may cause a negative RHS, which will not be informative. Moreover, Lemma 1 allows us to plug in any $u\in\mathcal{E}_0$ into the RHS of Equation (4), which is a key ingredient for the following main theorem of this section. Recall that $e_\nu(n,\hat{\Delta}_n) = \mathbb{E}[|\Delta_\nu - \hat{\Delta}_n|]$. By carefully choosing $\phi(n)$ and the instance u, we can derive the following worst-case bound for the error and the regret.

Theorem 1. *When $K=2$, for any admissible pair $(\pi,\hat{\Delta}_n)$, there always exists a hard instance $\nu\in\mathcal{E}_0$ that $e_\nu(n,\hat{\Delta}_n)\sqrt{\mathcal{R}_\nu(n,\pi)}$ is no less than a constant order: that is,*

$$\inf_{(\pi,\hat{\Delta}_n)} \max_{\nu\in\mathcal{E}_0} [e_\nu(n,\hat{\Delta}_n)\sqrt{\mathcal{R}_\nu(n,\pi)}] = \Omega(1).$$

Theorem 1 states that for any admissible pair $(\pi,\hat{\Delta}_n)$, there usually exists a challenging instance $\nu\in\mathcal{E}_0$ such that the product of estimation error and regret is lower bounded by a universal constant independent of n. This mathematically highlights the trade-off between the two objectives. A small regret will inevitably have a large error on the ATE estimation. Roughly speaking, the expected error is almost lower bounded by the inverse of the square root of the regret in the worst case $\left(\text{i.e., } e_\nu(n,\hat{\Delta}_n) = \Omega\left(1/\sqrt{\mathcal{R}_\nu(n,\pi)}\right)\right)$. In particular, because $\mathcal{R}_\nu(n,\pi) = \mathcal{O}(\log(n))$ for UCB and TS algorithms, no estimators cannot achieve smaller error than the order $\Omega\left(1/\sqrt{\log(n)}\right)$ consistently over all the possible instances. Although $\log(n)$ increases with n, the speed is rather slow, which explicitly shows the limitation of regret-optimal policies in terms of statistical power for estimating the ATE.

In order to prove Theorem 1, we construct two similar MAB instances ν_1 and ν_2 with the same distribution of the optimal arm but different treatment effects. This means that the distributions of the suboptimal arm are different. A high-quality ATE estimator must at least be able to distinguish between ν_1 and ν_2, which can only be achieved by playing the suboptimal arms. Thus, here comes the unavoidable regret. The detailed proof is delayed to Online Appendix B.1.

In Theorem 1, we have shown that no solution can perform better than a constant order in terms of $e_\nu(n,\hat{\Delta}_n)\sqrt{\mathcal{R}_\nu(n,\pi)}$ in the worst case. The following theorem states that one policy is Pareto optimal if it can achieve the constant order on $e_\nu(n,\hat{\Delta}_n)\sqrt{\mathcal{R}_\nu(n,\pi)}$ in terms of the dependence on n.

Theorem 2. *When $K=2$, an admissible pair $(\pi,\hat{\Delta})$ is Pareto optimal if it satisfies*

$$\max_{\nu\in\mathcal{E}_0} [e_\nu(n,\hat{\Delta})\sqrt{\mathcal{R}_\nu(n,\pi)}] = \tilde{\mathcal{O}}(1). \tag{5}$$

Together with Theorem 1, if Equation (5) is satisfied, we can directly draw the conclusion that $(\pi,\hat{\Delta})$ is optimal in terms of the metric $\max_{\nu\in\mathcal{E}_0}[e_\nu(n,\hat{\Delta})\sqrt{\mathcal{R}_\nu(n,\pi)}]$. However, Theorem 2 answers whether such optimality ensures our desired Pareto optimality. Comparing with the definition of Pareto optimality on the minimax multi-objective optimization problem, Equation (5) seems to be relatively more straightforward and practical to verify. For example, the traditional RCTs, where half of experimental units are assigned to the treatment group, are Pareto optimal with $e_\nu(n,\hat{\Delta}_{\mathrm{RCT}}) = \tilde{O}(1/\sqrt{n})$ and $\mathcal{R}_\nu(n,\pi_{\mathrm{RCT}}) = \Theta(n)$ for any $\nu\in\mathcal{E}_0$. This actually indicates that if the estimating error is the only measure we care about, then RCTs are just what we need.

2.2. An Optimal Design and a Necessary Condition

Although the RCTs are Pareto optimal, they may not allow for adjusting the balance between the two objectives. In this subsection, we propose an efficient and versatile experimental design satisfying the sufficient condition in Equation (5). Based on our proposed Pareto optimal design, we will prove that Condition (5) is actually necessary.

2.2.1. Algorithm and Regret Upper Bound. Here, we demonstrate the main ideas of the EXP3 with exploration (EXP3E) design whose pseudocode is provided in Algorithm 1. We adopt the famous EXP3 algorithm for adversarial MAB (see Auer et al. 2002a, Seldin et al. 2013) together with the idea to actively explore the suboptimal arm. Below, we illustrate the ideas in details.

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We first define a set of random variables $\hat{R}_t(a)$ for $a \in \{1,2\}$ and $t \in [n]$ based on inverse propensity score weight (IPW) estimator as

$$\hat{R}_t(a) = \hat{R}_{t-1}(a) + \frac{R_t}{\pi_t(a)}\mathbb{I}_{a=A_t} = \begin{cases} \hat{R}_{t-1}(a) + \frac{R_t}{\pi_t(a)} & \text{if } A_t = a; \\ \hat{R}_{t-1}(a) & \text{otherwise,} \end{cases}$$

where R_t is the reward collected at time t. Note that $\hat{R}_t(a)$ can provide an unbiased estimation of μ_a after being divided by t (i.e., $\mathbb{E}[\hat{R}_t(a)] = \mu_a t$). Thus, in line 14 in Algorithm 1, we use $\hat{R}_t(a)/t$ to estimate μ_a. One may think that a more straightforward way to estimate μ_a is the simple sample average $\frac{\sum_{s=1}^{t}\mathbb{I}_{a_s=A_t}R_s}{\sum_{s=1}^{t}\mathbb{I}_{a_s=A_t}}$. We are going to tackle the case where there exists some adversary in the following section. The simple sample average may lack robustness in adversarial settings. Additionally, within a fully stochastic environment, it is possible that the simple sample average is neither unbiased nor asymptotically normal. This deviation stems from the inherent correlation between the decision to take action a_t at time t and the past history, as demonstrated in prior studies (see, e.g., Xu et al. 2013, Luedtke and Van Der Laan 2016, Nie et al. 2018, Dimakopoulou et al. 2021, Hadad et al. 2021). Thus, the ATE estimator based on the simple sample averages could also be potentially biased. Additionally, we also define $\hat{R}_t^{\max} := \max_{a\in\{1,2\}}\hat{R}_t(a)$.

The first phase of our algorithm is aiming at identifying the best arm with well-controlled regret. In this phase, the algorithm is adaptively polishing its decision policy to gain confidence about which arm is the optimal one, according to the estimated reward $\hat{R}_t(a)$. There are many ways to map $\hat{R}_t(a)$ into probabilities, among which a popular choice is exponential weighting as follows:

$$\pi_t(a) = \frac{e^{\varepsilon_{t-1}\hat{R}_{t-1}(a)}}{\sum_{a\in\mathcal{A}} e^{\varepsilon_{t-1}\hat{R}_{t-1}(a)}}. \tag{6}$$

Note that the decision maker knows the exact $\pi_t(a)$, different from the classical offline ATE estimation where $\pi_t(a)$ could not be observed directly (see, e.g., Hirano et al. 2003, Kato et al. 2020). If at time t, there exists an arm a such that $\hat{R}_t(a)$ is larger than the other one by at least $\Omega(\sqrt{t})$, the algorithm believes that a is the optimal arm and eliminates the other arm. Formally, we use $\mathcal{A}_t$ to denote the set including all the candidate arms that may be the best. Our elimination rule is $\mathcal{A}_{t+1} = \mathcal{A}_t \backslash \{a \in \mathcal{A}_t : \hat{R}_t^{\max} - \hat{R}_t(a) > 2\sqrt{Ct}\}$ for some numerical constant C defined in Algorithm 1 (see line 13). Note that when the first phase ends is a stopping time with respect to the history $\mathcal{H}_t$. We define two stopping times as $\tau(a) = \max\{t : a \in \mathcal{A}_t\}$ for $a \in \{1,2\}$, and then, the first phase ends after $\min_{a\in\{1,2\}}\tau(a)$ periods. Define a good event $\mathcal{E}_1 := \{\text{The optimal arm is not eliminated}\}$.

Lemma 2. *Under the elimination rule, $\mathcal{E}_1$ occurs with probability at least $1 - 2n\delta$.*

Lemma 2 states that our elimination rule can always eliminate the suboptimal arm with high probability. Then, the next natural question is when the suboptimal arm will be eliminated (i.e., the magnitude of $\min_{a\in\{1,2\}}\tau(a)$), which is answered by the following Lemma 3.

Lemma 3. *If $\mathcal{E}_1$ happens, the first phase ends before $\frac{16C}{\Delta^2}+1$ periods with high probability: that is,*

$$\mathbb{P}\left(\mathcal{E}_1, \min_{a\in\{1,2\}}\tau(a) \geq \frac{16C}{\Delta^2}+1\right) \leq 2n\delta. \tag{7}$$

After eliminating the suboptimal arm, EXP3E operates into the second phase. The algorithm is forced to play the arm that was identified as the suboptimal one in the first phase with a carefully controlled probability $\alpha_t = 1/(2t^\alpha)$ (see lines 8 and 9 in Algorithm 1). α is an important input parameter that balances our two tasks. If $\alpha = 0$, then α_t remains to be $1/2$, meaning that EXP3E almost runs the RCTs in the second phase. One can expect a relatively large regret and strong statistical power. If $\alpha = 1$, α_t decays fast as t grows, which will not introduce much extra regret to the original EXP3 algorithm, and thus, EXP3E almost inherits the efficiency of the original EXP3 algorithm.

If the good event $\mathcal{E}_1$ happens, the accumulative regrets in the second phase can be controlled by $\mathbb{E}[\sum_{t=\min_{a\in\{1,2\}}\tau(a)}^{n}\mathbb{I}_{A_t\notin\mathcal{A}_t,\mathcal{E}_1}] \leq \sum_{t=1}^{n}\frac{1}{2t^\alpha} \leq \frac{n^{1-\alpha}}{2(1-\alpha)}\mathbb{I}_{\alpha<1} + \frac{1}{2}\log(n)\mathbb{I}_{\alpha=1}$, where the second inequality holds because $\sum_{t=1}^{n}\frac{1}{t^\alpha} \leq \int_0^n t^{-\alpha} = \frac{t^{1-\alpha}}{1-\alpha}$ for any $\alpha \in [0,1)$. When $\alpha = 1$, $\sum_{t=1}^{n}\frac{1}{2t} \leq \frac{1}{2}\int_0^n t^{-1} = \frac{\log(n)}{2}$. Together with Lemmas 2 and 3, we can derive the follows.

Algorithm 1 (EXP3 with Exploration for $K = 2$ (EXP3E))

1 **Input:** α and δ
2 **Initialization:** $\mathcal{A}_1 = \{1,2\}$, $\hat{R}_0(a) = 0$ for $a \in \{1,2\}$, $\varepsilon_0 = 0$, $C = 4(e^2+2)^2\left(\log\left(\frac{2}{\delta}\right)\right)^2$
3 **for** $t = 1,2,\ldots,n$, **do**
4 **if** $|\mathcal{A}_t| = 2$: //phase 1: best arm identification
5 $\varepsilon_t = \frac{1}{\sqrt{Ct}}$;
6 $\pi_t(1) = \frac{e^{\varepsilon_{t-1}\hat{R}_{t-1}(1)}}{e^{\varepsilon_{t-1}\hat{R}_{t-1}(1)}+e^{\varepsilon_{t-1}\hat{R}_{t-1}(2)}}$ and $\pi_t(2) = \frac{e^{\varepsilon_{t-1}\hat{R}_{t-1}(2)}}{e^{\varepsilon_{t-1}\hat{R}_{t-1}(1)}+e^{\varepsilon_{t-1}\hat{R}_{t-1}(2)}}$;
7 **else:** //phase 2: active exploration
8 $\alpha_t = \frac{1}{2t^\alpha}$;
9 $\pi_t(a) = 1 - \alpha_t$ if $a \in \mathcal{A}_t$; otherwise, $\pi_t(a) = \alpha_t$;
10 Select A_t according to π_t;
11 Observe reward R_t;
12 $\forall a \in \{1,2\}$: $\hat{R}_t(a) = \hat{R}_{t-1}(a) + \frac{R_t}{\pi_t(a)}\mathbb{I}_{a=A_t}$;
13 $\mathcal{A}_{t+1} = \mathcal{A}_t \backslash \{a \in \mathcal{A}_t : \hat{R}_t^{\max} - \hat{R}_t(a) > 2\sqrt{Ct}\}$;
14 **Output:** $\hat{\Delta}_t = \frac{1}{t}(\hat{R}_t(1) - \hat{R}_t(2))$;
15 **end for**

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Theorem 3. *When $K = 2$, let* EXP3E *run with any given $\alpha \in [0,1]$ and $\delta = 1/(2n^2)$. The regret is*

$$\mathcal{O}\left(\frac{\log(n)}{\Delta} + n^{1-\alpha}\Delta\log(n)\right). \tag{8}$$

The regret bound in Theorem 3 decreases with increasing α as a large α restricts the probability to play the suboptimal arm in the second phase. When $\alpha = 1$, Theorem 3 yields a regret upper bound of $\mathcal{O}\left(\frac{\log(n)}{\Delta} + \Delta\log(n)\right)$, which matches with the optimal regret bound of MAB in the current literature up to logarithmic factors. This means that if minimizing the accumulative regret is the only objective (i.e., ignoring the inference task), by setting $\alpha = 1$, the performance of our EXP3E is unimprovable in terms of the dependency on the experimental horizon n. On the other hand, when $\alpha = 0$, the regret upper bound grows linearly with the horizon n. To reiterate, when α is set to be zero, EXP3E is running RCTs as the exploration probability remains 1/2 in the second phase. Furthermore, because we assume Δ to be in a constant order (i.e., $|\Delta| = \Theta(1)$), Theorem 3 has an immediate corollary as follows.

Corollary 1. *With any given $\alpha \in [0,1]$, $\delta = 1/(2n^2)$ and $|\Delta| = \Theta(1)$, the regret of* EXP3E *is $\tilde{\mathcal{O}}(n^{1-\alpha})$.*

As we have mentioned, when $|\Delta| = \Theta(1)$, the optimal regret rate for MAB (without inference) is $\Theta(\log(n))$ (see, e.g., Lattimore and Szepesvári 2020), which shows that the upper bound in Corollary 1 is not rate optimal. We will later demonstrate the benefits of such suboptimality. The proofs of Lemmas 2 and 3 and Theorem 3 are deferred to Online Appendix B.

2.2.2. Inference for ATE. Now, we focus on estimating the treatment effect Δ. Because we have shown $\mathbb{E}[\hat{R}_t(a)] = \mu_a t$, we can define a set of martingales as $M_t^a = \hat{R}_t(a) - \mu_a t$ for $a \in \mathcal{A}$ and $M_t^{(1,2)} := M_t^1 - M_t^2 = t\hat{\Delta}_t - t\Delta$. Then, we can directly know that $\hat{\Delta}_t$ is an unbiased estimator of Δ (i.e., $\mathbb{E}[\hat{\Delta}_t] = \Delta$). For any $t \in [n]$, the martingale difference of $M_t^{(1,2)}$ can be bounded by $|M_t^{(1,2)} - M_{t-1}^{(1,2)}| = |\frac{R_t}{\pi_t(1)}\mathbb{I}_{A_t=1} - \frac{R_t}{\pi_t(2)}\mathbb{I}_{A_t=2} - \Delta| \le 2 + \frac{1}{\pi_t(1)} + \frac{1}{\pi_t(2)}$. Moreover, the variance of the martingale $M_t^{(1,2)}$ can be bounded as follows: $\sum_{t=1}^n \mathbb{E}\left[\left(\frac{R_t}{\pi_t(1)}\mathbb{I}_{A_t=1} - \frac{R_t}{\pi_t(2)}\mathbb{I}_{A_t=2} - \Delta\right)^2 | \mathcal{H}_{t-1}\right] \le \sum_{t=1}^n \frac{1}{\pi_t(1)} + \frac{1}{\pi_t(2)}$. A common concern about an IPW-based method is its large variance, especially when $\pi_t(a)$ is small (e.g., Dimakopoulou et al. 2021, Hadad et al. 2021). The blessing of online learning is that we can control the propensity score, but this also imposes an important challenge. Assigning a low probability to the suboptimal arm can reduce regret but increase the variance of estimators, potentially compromising the quality of inference. In our EXP3E, when $t > \min_{a\in\{1,2\}}\tau(a)$, $\frac{1}{\pi_t(1)} + \frac{1}{\pi_t(2)} \le 2t^\alpha + 2$ by the design of our EXP3E. When $t \le \min_{a\in\{1,2\}}\tau(a)$, from the proof of Theorem 3, our elimination rule contributes to securing $\frac{1}{\pi_t(a)} \le 1 + e^2$ for $a \in \mathcal{A}$. In this way, we successfully control the variance of $M_t^{(1,2)}$ to our desired level. By Bernstein's inequality (Freedman 1975), we can derive the following theorem.

Theorem 4. *If $K = 2$ and* EXP3E *runs with $\alpha \in [0,1]$ and $\delta < 2/e$, with probability at least $1-\delta$, for $t \in [n]$,*

$$|\hat{\Delta}_t - \Delta| \le \frac{(8e^2+16)\log\frac{2}{\delta}}{\sqrt{t^{1-\alpha}}}. \tag{9}$$

Furthermore, by taking $\delta = 1/(2n^2)$, $e(n,\hat{\Delta}) = \mathbb{E}[|\hat{\Delta}_n - \Delta|] = \tilde{\mathcal{O}}\left(1/\sqrt{n^{1-\alpha}}\right)$.

Although we adopt $e(n,\hat{\Delta})$ in our formulation to represent the statistical power of estimating the treatment effect, Equation (9) also enables the construction of valid confidence sequences, allowing for inference at any point when new data arrive. Confidence sequences consist of a sequence of confidence intervals that maintain uniform validity over time, commonly known as the anytime-valid inference (see, e.g., Ham et al. 2023). Formally, a sequence set of confidence intervals $\{V_t\}_{t\ge1}$ is considered a valid confidence sequence with a type 1 error of δ for the target parameter Δ if $\mathbb{P}(\forall t, \Delta \in V_t) \ge 1-\delta$. By reorganizing Equation (9), we observe that $\left\{\left[\hat{\Delta}_t - \frac{(8e^2+16)\log\frac{2}{\delta}}{\sqrt{t^{1-\alpha}}}, \hat{\Delta}t + \frac{(8e^2+16)\log\frac{2}{\delta}}{\sqrt{t^{1-\alpha}}}\right]\right\}_{t\ge1}$ can effectively serve as an anytime-valid confidence sequence: that is,

$$\mathbb{P}\left(\forall t, \Delta \in \left[\hat{\Delta}_t - \frac{(8e^2+16)\log\frac{2}{\delta}}{\sqrt{t^{1-\alpha}}}, \hat{\Delta}_t + \frac{(8e^2+16)\log\frac{2}{\delta}}{\sqrt{t^{1-\alpha}}}\right]\right) \ge 1-\delta. \tag{10}$$

For after-experiment inference (i.e., after observing all the data), the confidence interval that we can construct has a length in the order of $\tilde{\mathcal{O}}\left(1/\sqrt{n^{1-\alpha}}\right)$ given a desired confidence level δ. Additionally, Equation (10) can be adapted for hypothesis testing on the treatment effect being zero. Specifically, the null hypothesis is $H_0 : \Delta = 0$, and the alternative hypothesis is $H_1 : \Delta \ne 0$. We reject the null hypothesis when $\hat{\Delta}_t \notin \left[-\frac{(8e^2+16)\log\frac{2}{\delta}}{\sqrt{t^{1-\alpha}}}, \frac{(8e^2+16)\log\frac{2}{\delta}}{\sqrt{t^{1-\alpha}}}\right]$. It is important to note that the traditional t test may not be suitable in this context as the data in our experiment are adaptively collected and may not follow a Gaussian distribution.

Furthermore, different from most existing results on adaptive estimators in the causal inference literature that are asymptotic in nature, Theorem 4 is a finite sample property that has its own advantages when the number of samples is relatively small. Moreover, the RHS of Equation (9) is not dependent on the problem instance (i.e., Δ), making it applicable even for small $\Delta = \mathcal{O}(n^{-p})$. The error bound on the RHS of Equation (9) increases with the value of α. Intuitively, when α is small, EXP3E is more likely to explore during the second phase of our algorithm, improving the quality of

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inference. Specifically, if $\alpha = 0$, EXP3E is running RCTs during the second phase, which can be expected to provide strong statistical power, and Theorem 4 yields an $\tilde{\mathcal{O}}(1/\sqrt{n})$ error bound, recovering the results of RCTs. However, this comes at the cost of an increased regret bound, as seen in Theorem 3. Such observations illustrate the role that α plays in balancing the online decision making and adaptive inference. Together with Corollary 1 and Theorem 4, we can safely draw the following statement.

Corollary 2. *When $K = 2$, for any instance $\nu \in \mathcal{E}_0$ and $\alpha \in [0,1]$,* EXP3E *can guarantee $e_\nu(n,\hat{\Delta})\sqrt{\mathcal{R}_\nu(n,\pi)} = \tilde{\mathcal{O}}(1)$. Furthermore, by Theorem 1,* EXP3E *is Pareto optimal for any $\alpha \in [0,1]$.*

For simplicity, we denote the online decision-making policy and ATE estimator with input parameter α in Algorithm 1 as $(\pi_\alpha, \hat{\Delta}_\alpha)$. By Theorems 1 and 4 and Corollary 1, the front of $(\pi_\alpha, \hat{\Delta}_\alpha)$ is $\mathcal{F}(\pi_\alpha, \hat{\Delta}_\alpha) = \{(n^{1-\alpha}, 1/\sqrt{n^{1-\alpha}})\}$. Note that the front of $(\pi_\alpha, \hat{\Delta}_\alpha)$ is just a point. With different input of $\alpha \in [0,1]$, the fronts of our $(\pi_\alpha, \hat{\Delta}_\alpha)$ will cover the line $e(n,\hat{\Delta})\sqrt{\mathcal{R}_\nu(n,\pi)} \simeq 1$, which is the Pareto frontier, as shown in Figure 4. The front for $\alpha = 0$ coincides with the front of RCTs.

2.2.3. The Sufficient and Necessary Condition.

Based on the series of $(\pi_\alpha, \hat{\Delta}_\alpha)$, we can have the following theorem demonstrating the necessity of Condition (5). The main idea is that for any policy $(\tilde{\pi}, \tilde{\Delta})$ that violates Condition (5), there always exists an $\alpha \in [0,1]$ such that $(\pi_\alpha, \hat{\Delta}_\alpha)$ given by EXP3E will Pareto dominate $(\tilde{\pi}, \tilde{\Delta})$. The formal statement is as follows.

Theorem 5. *For $K = 2$, any Pareto optimal $(\pi^*, \hat{\Delta}^*)$ has $\max_{\nu \in \mathcal{E}_0}[e_\nu(n,\hat{\Delta}^*)\sqrt{\mathcal{R}_\nu(n,\pi^*)}] = \tilde{\mathcal{O}}(1)$.*

Together with Theorems 2 and 5, we formally confirm that the sufficient condition in Equation (5) is necessary for the Pareto optimal solutions.

Figure 4. (Color online) Fronts for $(\pi_\alpha, \hat{\Delta}_\alpha)$

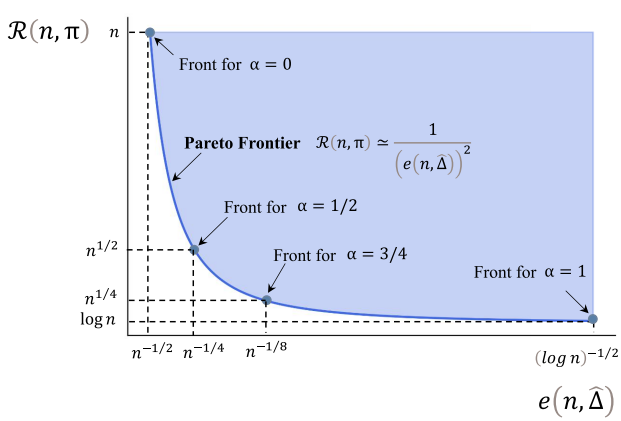
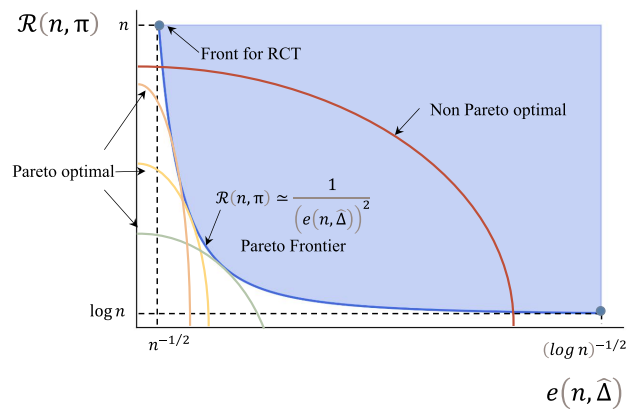


Figure 5. (Color online) Examples of Pareto (Non-)Optimal Solutions



Corollary 3. *When $K = 2$, the admissible pair $(\pi^*, \hat{\Delta}^*)$ is Pareto optimal to the minimax multi-objective optimization Problem (1) if and only if $\max_{\nu \in \mathcal{E}_0}[e_\nu(n,\hat{\Delta}^*)\sqrt{\mathcal{R}_\nu(n,\pi^*)}] = \tilde{\mathcal{O}}(1)$.*

In Figure 5, we present several examples of the possible fronts of some admissible policies. First, the lower bound (4) tells us that the front of any admissible policy has an intersection with the region $\mathcal{R}(n,\pi) \geq 1/(e(n,\hat{\Delta}))^2$ (colored blue in Figure 5). The sufficient and necessary condition indicates that the Pareto optimal solutions will only intersect with the blue region in Figure 5 on the boundary $\mathcal{R}(n,\pi) \simeq 1/(e(n,\hat{\Delta}))^2$. In turn, all of the policies whose intersections with the blue region in Figure 5 are only on the boundary are Pareto optimal. The red curve in Figure 5 is an example of the front of a non-Pareto optimal solution because it falls within the interior of the region.

3. Stochastic MAB Experiments with General K

In this section, we extend our model, algorithm, and analysis to general $K \geq 2$ arms. The main ideas of EXP3 with exploration for general K (EXP3EG) in Algorithm 2 follow from those of EXP3E. However, because the suboptimal arms are usually unlikely to be eliminated at the same time, EXP3EG cannot be divided into two phases explicitly anymore. Like before, we still maintain an active set $\mathcal{A}_t$ of the candidate arms that may be the best. For arm $a \notin \mathcal{A}_t$, we assign a time-varying but fixed probability $\alpha_t = 1/(Kt^\alpha)$ to be played. The probability of playing arm $a \in \mathcal{A}$ is determined by its estimated reward $\hat{R}_t(a)$ via a weighting method similar to Equation (6). Notably, EXP3EG can output $\hat{\Delta}^{(i,j)}$ for all $i \neq j \in [K]$ at the same time (line 10 in Algorithm 2), meaning that EXP3EG does not need to know which $\Delta^{(i,j)}$ is of interest in advance.

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Algorithm 2 (EXP3 with Exploration for General K)

Input: α and δ
Initialization: $\mathcal{A}_1 = \{1,2,\ldots,K\}$, $\hat{R}_0(a) = 0$ for $a \in \{1,2,\ldots,K\}$, $C = (4K^2(e^2+1)+2)^2\left(\log\left(\frac{2}{\delta}\right)\right)^2$
for $t = 1,2,\ldots,n$, **do**
  $\varepsilon_t = \frac{1}{\sqrt{Ct}}$ and $\alpha_t = \frac{1}{Kt^{\alpha}}$;
  $\forall a \in \mathcal{A}_t$: $\pi_t(a) = (1-|\mathcal{A}_t^c|\alpha_t)\frac{e^{\varepsilon_{t-1}\hat{R}_{t-1}(a)}}{\sum_{a'\in\mathcal{A}_t} e^{\varepsilon_{t-1}\hat{R}_{t-1}(a')}}$; $\forall a \notin \mathcal{A}_t$: $\pi_t(a) = \alpha_t$;
  Select A_t according to π_t;
  Observe reward R_t;
  $\forall a \in \{1,2,\ldots,K\}$: $\hat{R}_t(a) = \hat{R}_{t-1}(a) + \frac{R_t}{\pi_t(a)}\mathbb{I}_{a=A_t}$;
  $\mathcal{A}_{t+1} = \mathcal{A}_t \backslash \{a : \hat{R}_t^{\max} - \hat{R}_t(a) > 2\sqrt{Ct}\}$;
  Output: For $i,j \in [K]$ and $i \neq j$: $\hat{\Delta}_t^{(i,j)} = \frac{1}{t}(\hat{R}_t(i) - \hat{R}_t(j))$;
end for

Similarly, we define a set of stopping times $\tau(a) = \max\{t : a \in \mathcal{A}_t\}$ for all $a \in [K]$, which are the time periods after which arm a is eliminated, and the good event $\mathcal{E}_1 :$ = {The optimal arm is not eliminated}. With the union bound and Lemma 2, we can know $\mathbb{P}(\mathcal{E}_1) \geq 1 - 2Kn\delta$. For a fixed suboptimal arm a, Lemma 3 still holds (i.e., $\mathbb{P}\left(\mathcal{E}_1, \tau(a) \geq \frac{16C}{\Delta(a)^2}+1\right) \leq 2n\delta$). Then, we can derive $\mathbb{P}\left(\mathcal{E}_1, \left\{\exists a \neq a^*, \tau(a) \geq \frac{16C}{\Delta(a)^2}+1\right\}\right) \leq 2Kn\delta$ with the union bound again. Therefore, the regret of EXP3EG can be bounded as follows, and the detailed proof is provided.

Theorem 6. *Let* EXP3EG *run with* $\alpha \in [0,1]$ *and* $\delta = 1/2n^2$. *The regret is* $\mathcal{O}\left(\sum_{a\in[K]/\{a^*\}} \frac{\log(n)}{\Delta(a)} + \Delta(a)n^{1-\alpha}\log(n)\right)$.

When $\alpha = 1$, the regret upper bound in Theorem 6 matches with the minimax lower bounds for MAB problems up to a logarithmic factor. Also, because we care about $|\Delta(a)| = \Theta(1)$, the regret bounds becomes $\tilde{\mathcal{O}}(n^{1-\alpha})$. For inference, following Section 2.2, we introduce a series of martingales $M_t^{(i,j)} := M_t^i - M_t^j$ for any $i \neq j \in [K]$, where $M_t^i = \hat{R}_t(i) - \mu_i t$ and $M_t^{(i,j)} = t(\hat{\Delta}_n^{(i,j)} - \Delta^{(i,j)})$. An immediate result is the unbiasedness (i.e., $\mathbb{E}[\hat{\Delta}_n^{(i,j)}] = \Delta^{(i,j)}$). We extend Theorem 4 as follows.

Theorem 7. *If* EXP3EG *runs with* $\alpha \in [0,1]$, *for any fixed* $i \neq j \in [k]$, *and* $\delta < 2/e$, *with probability at least* $1-\delta$, *for all* $t \in [n]$,

$$|\hat{\Delta}_t^{(i,j)} - \Delta^{(i,j)}| \leq \frac{4(2+2K^2(1+e^2))\log\frac{2}{\delta}}{\sqrt{t^{1-\alpha}}}. \tag{11}$$

Particularly, after taking $\delta = 1/(2n^2)$, $\max_{i<j\leq K} e(n, \hat{\Delta}_n^{(i,j)}) = \mathcal{O}\left(1/\sqrt{n^{1-\alpha}}\right)$.

Note that the estimation error of ATE is uniformly bounded by the RHS of Equation (11) for any $i \neq j$ with high probability. Thus, we can have $\max_{i<j\leq K} e(n, \hat{\Delta}_n^{(i,j)}) = \mathcal{O}\left(1/\sqrt{n^{1-\alpha}}\right)$. Then, we can naturally derive $(\max_{i<j\leq K} e_\nu(n, \hat{\Delta}_n^{(i,j)}))\sqrt{\mathcal{R}_\nu(n,\pi)} = \tilde{O}(1)$ for any $\nu \in \mathcal{E}_0$. Taking i^* and j^* to be the best arm and the second-best arm, respectively, we can always have $\max_{i<j\leq K} e(n, \hat{\Delta}_n^{(i,j)}) \geq e(n, \hat{\Delta}_n^{(i^*,j^*)})$. Then, by such an observation, we reduce the problem with $K \geq 2$ to $K = 2$. Thus, Lemma 1 and Theorem 2 can be generalized to here, and accordingly, the sufficient condition for Pareto optimal with $K > 2$ becomes $\max_{\nu\in\mathcal{E}_0}(\max_{i<j\leq K} e_\nu(n, \hat{\Delta}_n^{(i,j)}))\sqrt{\mathcal{R}_\nu(n,\pi)} = \tilde{O}(1)$. Hence, EXP3EG is Pareto optimal for all $\alpha \in [0,1]$. By such a claim, the necessity of the condition follows similarly from Theorem 5. We can naturally extend Corollary 3 to $K \geq 2$ as follows.

Theorem 8. *The admissible pair* $(\pi^*, \hat{\Delta}^*)$ *is Pareto optimal to the minimax multi-objective optimization Problem* (1) *if and only if* $\max_{\nu\in\mathcal{E}_0}[(\max_{i<j\leq K} e_\nu(n, \hat{\Delta}_n^{*(i,j)}))\sqrt{\mathcal{R}_\nu(n,\pi^*)}] = \tilde{O}(1)$.

Remark 1. At times, experimenters may only be interested in estimating the gap between two specific arms, such as the best and second-best arms. In such cases, they might consider $\max_{i<j\leq K} e_\nu(n, \hat{\Delta}_n^{(i,j)})$ as a too conservative metric. Instead, they may prefer to focus on a specific pair (i, j) and examine $\Delta^{(i,j)}$. In order to tackle such a concern, we establish a stronger lower bound that indicates that the difficulty of estimating a specific $\Delta^{(i,j)}$ cannot be significantly reduced, even if we are willing to sacrifice the accuracy of estimating all other gaps. Formally, we derive the following lower bound.

Theorem 9. *For any given pair* (i, j) *and any admissible pair* $(\pi, \hat{\Delta}_n)$, *there always exists a hard instance* $\nu \in \mathcal{E}_0$ *that* $e_\nu(n, \hat{\Delta}_n^{(i,j)})\sqrt{\mathcal{R}_\nu(n,\pi)}$ *is no less than a constant order: that is,*

$$\inf_{(\pi,\hat{\Delta}_n)} \max_{\nu\in\mathcal{E}_0}[e_\nu(n, \hat{\Delta}_n^{(i,j)})\sqrt{\mathcal{R}_\nu(n,\pi)}] = \Omega(1).$$

Theorem 9 indicates that $e_\nu(n, \hat{\Delta}_n^{(i,j)})$ is always lower bounded by $\Omega\left(1/\sqrt{\mathcal{R}_\nu(n,\pi)}\right)$ for any $i \neq j$. Specifically, for a policy $\tilde{\pi}$ that ensures $\mathcal{R}_\nu(n,\tilde{\pi}) = \tilde{\mathcal{O}}(n^\beta)$, even when focusing solely on estimating $\Delta^{(i,j)}$, $e_\nu(n, \hat{\Delta}_n^{(i,j)})$ cannot be smaller than $\Omega(n^{-\beta/2})$ in the worst case. This conveys an important message that the difficulty of estimation remains nearly identical, regardless of whether we are concerned with estimating a specific gap or all the gaps simultaneously, as the estimation error's order remains consistent. This implies the possibility of all $K(K-1)/2$ estimation errors being in the order of $\tilde{\Theta}(n^{-\beta/2})$. Building upon Theorem 9, the sufficient and necessary condition in Theorem 8 remains valid when replacing $\max_{i<j\leq K} e_\nu(n, \hat{\Delta}_n^{(i,j)})$ with $e_\nu(n, \hat{\Delta}_n^{(i,j)})$ for a specific pair (i, j).

4. MAB Experiments with Adversarial Baseline Rewards

In practice, the assumption that the rewards for one treatment are i.i.d. generated may fail to hold. For example, in the application of mobile health (mHealth), Greenewald et al. (2017) mention that the treatment

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effect can be well formulated to be stationary over time, but the observed rewards heavily depend on human behaviors, which are very hard to model using fully stochastic and stationary models. On the other hand, the adversarial MAB models are at the opposite extreme. If the treatment effect is chosen by an adversary arbitrarily, ATE is no longer a meaningful metric to infer. Moreover, a commonly held conviction is that treatment effects may be more structured than the outcomes themselves (see Kennedy 2020, Carranza et al. 2023). In this section, we consider an environment where an adversary exists. Specifically, the decision maker chooses an action $a_t \in \mathcal{A}$ and receives rewards

$$r_t(a_t) = \mu_{a_t} + f_t + \varepsilon_t, \tag{12}$$

where μ_{a_t} is an unknown parameter for each arm, ε_t is an unobservable zero-mean random variable, and f_t is chosen by an adaptive adversary at the beginning of round t. f_t does not depend on the selected action a_t. Thus, we call f_t the *adversarial baseline rewards*. Such a reward structure described in Model (12) is gaining attention in various fields, including mHealth (Greenewald et al. 2017) and bandits (Krishnamurthy et al. 2018, Kim and Paik 2019). We also assume $r_t(a) \in [-1,1]$, for all $a \in \mathcal{A}$. We do not impose any other structure assumptions on f_t. In this setting, we need to redefine the history as $\mathcal{H}_t := (f_1, a_1, r_1, \ldots, f_t, a_t, r_t)$. Despite the existence of the adversarial f_t, the optimal action $a^* = \arg\max_{a\in\mathcal{A}} \mu_a$ and the regret $\mathcal{R}(n,\pi)$ remain unchanged. The treatment effects $\Delta^{(i,j)} = \mu_{a_i} - \mu_{a_j}$ also remain, and the challenge lies in the fact that at each t, we can only observe $r_t(a_t)$ and not the value of f_t.

Although EXP3EG was initially designed for stochastic MAB experiments, we will show that it also works well with the reward structure (12). Technically, the first challenge is that the presence of the adversarial f_t makes $M_t^a := \hat{R}_t(a) - \mu_a t$ no longer a martingale because $\mathbb{E}[M_t^a \mid \mathcal{H}_{t-1}] = M_{t-1}^a + f_t$. Intuitively, because f_t can change arbitrarily, we virtually cannot estimate μ_a without explicit observations of f_t. However, fortunately, $M_t^{(i,j)} := M_t^i - M_t^j$ remains to be a martingale because the adversarial part can be cancelled out (i.e., $\mathbb{E}[M_t^{(i,j)} \mid \mathcal{H}_{t-1}] = M_{t-1}^i + f_t - M_{t-1}^j - f_t = M_{t-1}^{(i,j)}$). Thus, the advantage of IPW-based algorithms is further demonstrated compared with the algorithms based on the simple sample averages, like UCB, because it can help diminish the effect of f_t. As a contrast, for the algorithms based on the simple sample averages, the adversarial f_t will introduce extra bias. Specifically, $\frac{\sum_{s=1}^t \mathbb{I}_{a_s=i} f_s}{\sum_{s=1}^t \mathbb{I}_{a_s=i}} - \frac{\sum_{s=1}^t \mathbb{I}_{a_s=j} f_s}{\sum_{s=1}^t \mathbb{I}_{a_s=j}}$ can be any value in $[-1,1]$, which will be an extra bias to the ATE estimator. Additionally, from line 5 in Algorithm 2, we can know that the probability of playing an action a in $\mathcal{A}_t$ is proportional to $\frac{e^{\varepsilon_{t-1}\hat{R}_{t-1}(a)}}{\sum_{a'\in\mathcal{A}_t} e^{\varepsilon_{t-1}\hat{R}_{t-1}(a')}} = \frac{1}{\sum_{a'\in\mathcal{A}_t} e^{\varepsilon_{t-1}(\hat{R}_{t-1}(a')-\hat{R}_{t-1}(a))}}$, which intuitively illustrates that careful control of the difference of $\hat{R}_{t-1}(a')$ and $\hat{R}_{t-1}(a)$ can also be enough for EXP3EG to work, even if $\hat{R}_{t-1}(a)$ performs adversarially. Therefore, although we can no longer have an unbiased estimations on μ_a for each arm, $\hat{\Delta}_n^{(i,j)}$ is still an unbiased estimator of $\Delta^{(i,j)}$, for any $i \neq j \in [K]$, as shown in the following theorem.

Theorem 10. *If* EXP3EG *runs with* $\alpha \in [0,1]$ *and* $\delta < 2/e$ *under Model* (12), *for any* $i,j \in [K]$ *and* $i \neq j$, $\mathbb{E}[\hat{\Delta}_t^{(i,j)}] = \Delta^{(i,j)}$. *Moreover, with probability at least* $1-\delta$, *for all* $t \in [n]$,

$$|\hat{\Delta}_t^{(i,j)} - \Delta^{(i,j)}| \leq \frac{4(2+2K^2(1+e^2))\log\frac{2}{\delta}}{\sqrt{t^{1-\alpha}}}.$$

Particularly, after taking $\delta = \frac{1}{2n^2}$, $\max_{i<j\leq K} e(n, \hat{\Delta}_n^{(i,j)}) = \mathcal{O}\left(\frac{1}{\sqrt{n^{1-\alpha}}}\right)$.

For the regret, the good event $\mathcal{E}_1$ is guaranteed to happen with probability $1-2Kn\delta$ (i.e., $\mathbb{P}(\mathcal{E}_1) \geq 1-2Kn\delta$). However, the proof is different because M_t^a is no longer a martingale, which is integrated in the detailed proof for Theorem 11 in Online Appendix D.1. We now can have the bound for EXP3EG when the reward structures follow from Equation (12).

Theorem 11. *Under model* (12), *let* EXP3EG *run with* $\alpha \in [0,1]$ *and* $\delta = \frac{1}{2n^2}$. *The regret is* $\mathcal{O}\left(\sum_{a\in[K]/\{a^*\}} \frac{\log(n)}{\Delta(a)} + \Delta(a)\, n^{1-\alpha}\right)$.

Theorems 10 and 11 fully recover the results in Section 3. Moreover, note that when f_t is a constant over time, Model (12) reduces to the stochastic MAB experiments studied in Section 3. The lower-bound result in Theorem 1 can directly serve as a lower bound here. Thus, EXP3EG achieves the Pareto optimality, showing its robustness to the adversarial baseline reward.

Remark 2 (Fully Adversarial Setting). As we have previously mentioned, in most cases, the purpose of conducting experiments is to gain generalizable knowledge, such as the average treatment effect, for future use. In a fully adversarial environment, there is little information that can be carried forward to the future. However, recently, there is a growing interest in inferring the treatment effect solely during the experiment period from a retrospective view without considering its external validity (see, e.g., Ham et al. 2023). In this context, we can redefine our estimands, and our previous results still hold. During each time step $t \leq n$, the environment generates a reward $r_t(a) \in [-1,1]$ for each arm $a \in \mathcal{A}$ based on a distribution $P_{a,t}$, where we allow $P_{a,t}$ to be adversarially chosen for different t. Accordingly, we define $\mu_{a,t} := \mathbb{E}[r_t(a)]$. Subsequently, the expected average outcome of arm a can be updated as $\mu_a = \frac{1}{n}\sum_{t=1}^n \mu_{a,t}$, and our estimand becomes $\Delta^{(i,j)} = \mu_i - \mu_j$ accordingly. If the optimal arm consistently remains the same, denoted as a^*, over time, with $\mu_{a^*,t} - \mu_{a,t} = \Theta(1)$ for $a \neq a^*$, EXP3EG retains the desired properties derived

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earlier without requiring any modifications. This means that we allow the rewards of arms to be adversarially chosen, but the best arm should always be fixed. We do not impose restrictions on the rank of all other arms expect the fixed best arm. The rationale behind requiring a fixed best arm lies in the elimination-based design of our approach. The elimination strategy is meaningful only when the best arm is fixed. Otherwise, an eliminated arm might perform exceptionally well after elimination in an adversarial scenario. If the best arm is not fixed, the regret analysis loses its validity, but our estimator can still maintain an $\mathcal{O}(1/\sqrt{n^{1-\alpha}})$ error bound. Formal technical theorems and proofs are available in Online Appendix D.3.

5. Batched MAB Experiments

In many applications, experiments must be conducted in batches. For instance, in clinical trials, groups of patients are treated simultaneously. Observing outcomes, like five-year survival rates or vaccine effectiveness immediately after treatment, is not possible. Instead, the decision maker receives the delayed outcomes in the batch form and can only update its policy at the start of each batch based on the information collected from previous batches. Therefore, our fully adaptive framework and design in the preceding may not be applicable in this scenario.

Formally, for a batched MAB experiment, the entire experimental horizon n is split into M batches represented by a *grid* $\mathcal{T} = \{t_1, \ldots, t_M\}$ and $1 \le t_1 < t_2 < \cdots < t_M = n$. The number of batches, M, is usually predetermined, based on regulations or the cost of patient recruitment, and given as an input. The admissible policy π_t at a time $t_{j-1} < t \le t_j$ can only depend on the history up to time t_{j-1}, meaning that π_t can only use information collected strictly anterior to the current batch. Accordingly, the estimator $\hat{\Delta}_t^{(i,j)}$ can only be based on $H_{t_{j-1}}$ as well. The remaining components of the formulation, including the instance class $\mathcal{E}_0$, the regret definition $\mathcal{R}(n,\pi)$, and the error $e(t,\hat{\Delta})$, remain the same as in Section 1.1. Such a batched setting has received significant attention in the recent literature (see, e.g., Perchet et al. 2016, Gao et al. 2019, Zhang et al. 2020).

Algorithm 3 (Batched Successive Elimination Design (BaSED))

1 **Input:** $T, M, \mathcal{T} = \{t_1, \ldots, t_M\}, \alpha$
2 **Initialization:** $\mathcal{A}_1 = \{1, 2, \ldots, K\}$
3 **for** $m = 1, 2, \ldots, M-1$, **do**
4 (a) During $[t_{m-1}+1, t_m]$, pull each arm $a \in \mathcal{A}_{t_{m-1}+1}$ for a same number of times;
5 (b) At time t_m:
6 Calculate $\hat{R}_{t_m}(a) = \frac{\sum_{s=1}^{t_m} R_s \mathbb{I}_{a=A_s}}{\sum_{s=1}^{t_m} \mathbb{I}_{a=A_s}}$ for all $a \in [K]$;
7 $\hat{R}_{t_m}^{\max} = \arg\max_{a \in \mathcal{A}_{t_{m-1}+1}} \hat{R}_{t_m}(a)$;
8 $\tau_m = \sum_{s=1}^{t_m} \mathbb{I}_{a=A_s}$ for any $a \in \mathcal{A}_{t_{m-1}+1}$;
9 $\mathcal{A}_{t_m+1} = \mathcal{A}_{t_{m-1}+1} \backslash \{a : \hat{R}_{t_m}^{\max} - \hat{R}_{t_m}(a) > \sqrt{12\log(nK)/\tau_m}\}$;
10 **end for**
11 During $[t_{M-1}+1, n]$,
12 pull arm $i_0 = \arg\max_{a \in \mathcal{A}_{t_{M-1}+1}} \hat{R}_{t_{M-1}}(a)$ exactly $(n - t_{M-1}) - (K-1)\lceil \frac{(n-t_{M-1})}{Kn^{\alpha}} \rceil$ times;
13 for any $i \in [K] \backslash \{i_0\}$, pull arm i $\lceil \frac{(n-t_{M-1})}{Kn^{\alpha}} \rceil$ times.
14 **if** $\alpha \le 1 - \frac{1}{M}$, **do**
15 Calculate $\hat{R}_n(a) = \frac{\sum_{s=t_{M-1}+1}^{n} R_s \mathbb{I}_{a=A_s}}{\sum_{s=t_{M-1}+1}^{n} \mathbb{I}_{a=A_s}}$ for all $a \in [K]$;
16 **else**
17 Calculate $\hat{R}_n(a) = \frac{\sum_{s=1}^{t_1} R_s \mathbb{I}_{a=A_s}}{\sum_{s=1}^{t_1} \mathbb{I}_{a=A_s}}$ for all $a \in [K]$.
18 **Output:** For $i, j \in [K]$ and $i \ne j$: $\hat{\Delta}_n^{(i,j)} = (\hat{R}_n(i) - \hat{R}_n(j))$.

Compared with the stochastic MAB experiments in the preceding, the batched MAB experiment requires the design of $\mathcal{T}$, which splits the entire experimental horizon n into M batches. If $M = 1$, the setting resembles RCTs, where the decision maker decides its static policy even before the experiments begin. On the other hand, $M = n$ recovers the traditional stochastic MAB setting, where the decision maker can immediately observe the outcome and adjust its policy at each time period. In the current literature, Gao et al. (2019) show that the best achievable regret rate under the instance class $\mathcal{E}_0$ is $\tilde{\Theta}(n^{\frac{1}{M}})$. They propose an efficient rate-optimal algorithm based on successive elimination, which our design (BaSED) also leverages. The details of BaSED are presented in Algorithm 3. Specifically, for any given batch M, following the design of Perchet et al. (2016) and Gao et al. (2019), the grid is defined to be

$$u_1 = c_b n^{\frac{1}{M}}, \qquad u_m = u_1^m, \qquad t_m = \lfloor u_m \rfloor, \qquad m \in [M],$$

where c_b is a numerical constant that guarantees $t_M = n$. In essence, we design t_m to be $\Theta(n^{m/M})$, with the length of each batch increasing almost geometrically. Like EXP3EG, BaSED maintains an active set $\mathcal{A}_{t_m+1}$ for each batch, which includes all potential best arms. In each small batch, BaSED plays every arm in $\mathcal{A}_{t_m+1}$ for the same number of times (line 4 in Algorithm 3). Thus, the number of times being played (i.e., τ_m in line 8 in Algorithm 3) is the same for any arm in the active set. Different from EXP3EG, the estimators of rewards that BaSED adopts during the first $M-1$ batches are just the simple sample average (line 6 in Algorithm 3). One of the reasons is that the data collected within a small batch can be seen as i.i.d. generated, which is the blessing of the batch setting. Moreover, given all of the observations of the previous batches, the number of times that each arm will be played in the current batch is deterministic. An arm will be eliminated from the active set if its estimated reward is too low compared with the highest one (line 9 in Algorithm 3). In the last batch, which is also the longest of length $\Theta(T)$, BaSED plays the one with the

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largest estimated reward in the active set $\Theta(T - T^{1-\alpha} - T^{(M-1)/M}) = \Theta(T)$ times, whereas all the other arms are forced to be played $\Theta(T^{1-\alpha})$ times. Another observation is that during the first batch, every arm is played $\Theta(T^{1/M})$ times independently. The quality of the Δ estimate is largely determined by the arm played the least; so, when $\alpha \le 1 - (1/M)$, BaSED relies only on data from the last batch (line 15 in Algorithm 3), whereas it uses data from the first batch (line 17 in Algorithm 3) when $\alpha > 1 - (1/M)$. Admittedly, this design may be sample inefficient because many data points are discarded. It bring us much convenience in analyzing the quality of estimates because the data are i.i.d. We will also show that it is sufficient for a rate-optimal estimator. The main theorem of this section is as follows.

Theorem 12. *Let* BaSED *run with* $\alpha \in [0,1]$. *For any* $\nu \in \mathcal{E}_0$,

- *the regret* $\mathcal{R}(n,\pi) = \tilde{\mathcal{O}}(n^{1/M} \vee n^{1-\alpha})$;
- *the estimated error* $\max_{i<j\le K} e(n, \hat{\Delta}_n^{(i,j)}) = \tilde{\mathcal{O}}\left(1/\sqrt{n^{1/M} \vee n^{1-\alpha}}\right)$.

The lower bound established in Theorem 1 holds for any policy π and estimator $\hat{\Delta}_n^{(i,j)}$, regardless of whether the policy is in a batch fashion. Therefore, we can conclude that BaSED is Pareto optimal. Also, note that α only starts to take effect when $\alpha \le 1 - (1/M)$. The best regret rate that can be achieved by manipulating α is in the order of $n^{\frac{1}{M}}$, which is the optimal regret rate with M batches shown by Perchet et al. (2016) and Gao et al. (2019). This implies that there is no incentive that any decision maker will choose a large $\alpha > 1 - (1/M)$. The results for the entire $\alpha \in [0,1]$ in Theorem 12 are presented for completeness. Moreover, when $M = \Omega(\log n)$, the $n^{1-\alpha}$ term always dominates, and Theorem 12 recovers the results in Section 3, indicating that $\log n$ batches are sufficient to provide competitive theoretical performances compared with the fully dynamic case. Similar observations can also be found in the field of bandits with switching constraints (see, e.g., Simchi-Levi and Xu 2023).

As the data from a single arm within one batch can be considered i.i.d., we can apply standard techniques to establish the central limit theorem for the estimator employed in BaSED. This represents an advantage conferred by the batch-based algorithm, distinguishing it from the fully adaptive algorithms discussed earlier.

Proposition 1. *As* $n \to \infty$,

$$\sqrt{n(a)}\frac{\hat{R}_n(a) - \mu_a}{s_n(a)} \to \mathcal{N}(0,1),$$

where $n(a)$ *is the number of samples* $\hat{R}_n(a)$ *used and* $s_n(a)$ *is any consistent estimator of the standard deviation of arm* a. *To be more specific, if* $\alpha \le 1 - (1/M)$, $n(a) = \Omega(n^{1-\alpha})$ *and* $s_n(a) = \sqrt{\frac{\sum_{s=t_{M-1}+1}^{n}(R_s\mathbb{I}_{a=A_s} - \hat{R}_n(a))^2}{\sum_{s=t_{M-1}+1}^{n}\mathbb{I}_{a=A_s}}}$. *If* $\alpha > 1 - (1/M)$, $n(a) = \Omega(n^{1/M})$ *and* $s_n(a) = \sqrt{\frac{\sum_{s=1}^{t_1}(R_s\mathbb{I}_{a=A_s} - \hat{R}_n(a))^2}{\sum_{s=1}^{t_1}\mathbb{I}_{a=A_s}}}$.

As previously discussed, our estimator achieves rate optimality, although we only use the data from one batch. Although incorporating data from different batches may enhance the constant dependence, it does so at the cost of sacrificing the i.i.d. property of the data, which introduces extra and nontrivial technical challenges (see, e.g., Zhang et al. 2020).

Remark 3. By appropriately choosing M, BaSED can enjoy similar theoretical guarantees as EXP3EG in terms of dependence on n, and it can also induce an asymptotic normality property. However, certain considerations may limit the universal adoption of BaSED. Theoretically, unlike EXP3EG, batch-based designs often face challenges in simultaneously achieving optimality for worst-case and instance-dependent scenarios. Specifically, in order to achieve the worst-case and instance-dependent regret optimality, different grid sizes are typically required (refer to Perchet et al. 2016 and Gao et al. 2019). For instance-dependent optimality, the grid is designed as an exponential sequence of $n^{1/M}$ as we employ here. On the other hand, for worst-case optimality, the grid needs to be chosen as $\Theta\left(n^{\frac{2-2^{1-k}}{2-2^{1-M}}}\right)$ for the kth batch. In contrast, EXP3EG eliminates the need for algorithm modification to achieve optimality for both worst-case and instance-dependent scenarios, as discussed in the next section. Furthermore, batch-based algorithm grid designs heavily rely on prior knowledge of n, which is practical in many settings but lacks flexibility if the experimenter is uncertain about the number of experimental units at the beginning or receives additional resources midexperiment. EXP3EG, in comparison, only depends on n through the choice of δ, allowing for a conservative selection to accommodate uncertainty in n. Additionally, despite theoretical similarities, batch-based algorithms tend to exhibit less favorable empirical performance than fully adaptive algorithms in terms of regret. The intuition lies in the fully adaptive algorithm's flexibility to eliminate suboptimal arms whenever it gains confidence, leading to smaller empirical regrets. This observation is supported by numerical studies, such as those conducted by Gao et al. (2019). Real-world industrial practice also sounds a cautionary note against adopting batching: for example, Amazon's experience as highlighted by Fiez et al. (2022).

6. MAB Experiments Without the Large-Gap Assumption

In the previous sections, we considered the instance class $\mathcal{E}_0$, where $\Delta^{(i,j)} = \Theta(1)$ for all $i,j \in [K]$. This class,

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known as the "large-gap" or "well-separated" instance class (see, e.g., Kalvit and Zeevi 2024), allows us to treat $\Delta^{(i,j)}$ as a constant independent of n and disregard its influence when establishing the necessary and sufficient condition for Pareto optimality. In this section, we extend our analysis beyond the large-gap assumption and demonstrate that our EXP3E and EXP3EG designs remain Pareto optimal. To simplify the discussion, we focus on the case of $K = 2$ and EXP3E as the extension to $K > 2$ and EXP3EG follows naturally. We define the instance class without the assumption $\Delta = \Theta(1)$ as $\tilde{\mathcal{E}}_0$, where it is evident that $\mathcal{E}_0 \subset \tilde{\mathcal{E}}_0$.

The first fundamental observation is that the lower bound established in Theorem 1 can directly serve as a lower bound here because $\mathcal{E}_0 \subset \tilde{\mathcal{E}}_0$. Specifically,

$$\inf_{(\pi,\hat{\Delta}_n)} \max_{\nu\in\tilde{\mathcal{E}}_0} [e_\nu(n,\hat{\Delta}_n^{(i,j)})\sqrt{\mathcal{R}_\nu(n,\pi)}] \geq \inf_{(\pi,\hat{\Delta}_n)} \max_{\nu\in\mathcal{E}_0} [e_\nu(n,\hat{\Delta}_n^{(i,j)})\sqrt{\mathcal{R}_\nu(n,\pi)}] = \Omega(1).$$

Therefore, we can extend the claim of the sufficiency of our condition to the setting without the large-gap assumption. This means that the quantity of our interest above (i.e., $e_\nu(n,\hat{\Delta})\sqrt{\mathcal{R}_\nu(n,\pi)}$) is still informative enough to indicate the Pareto optimality under the general instance class $\tilde{\mathcal{E}}_0$.

Corollary 4 (Sufficient Condition). *When $K = 2$, if an admissible pair $(\pi,\hat{\Delta})$ satisfies*

$$\max_{\nu\in\tilde{\mathcal{E}}_0} \left[e_\nu(n,\hat{\Delta})\sqrt{\mathcal{R}_\nu(n,\pi)}\right] = \tilde{\mathcal{O}}(1),$$

it is Pareto optimal.

Now, we are shifting our gear to investigate the performance of EXP3E when facing the instance class $\tilde{\mathcal{E}}_0$. One key claim that we make is that in the absence of the large-gap assumption, it is unrealistic to expect a regret upper bound of $\log(n)$ under worst-case scenarios. The existing literature extensively explores the minimax lower bound for regret, which has been well established as $\Omega(\sqrt{n})$. Specifically, it has been shown that $\inf_\pi \max_{\nu\in\tilde{\mathcal{E}}_0} \mathcal{R}_\nu(n,\ \pi) = \Omega(\sqrt{n})$. An important observation is that the instance-dependent regret bound for EXP3E, as stated in Theorem 3, remains valid even without the $\Delta = \Theta(1)$ assumption. Additionally, a trivial upper bound for regret is $n\Delta$. Based on these insights, we can derive the following corollary on the instance-independent upper bound.

Corollary 5. *Let* EXP3E *run with any given $\alpha \in [0,1]$ and $\delta = 1/(2n^2)$. The regret is $\tilde{\mathcal{O}}(\sqrt{n} \vee n^{1-\alpha})$.*

The first term of $\sqrt{n}$ aligns with the previously mentioned lower bound for worst-case regret and cannot be eliminated. Only when $\alpha < 1/2$ does the second term begin to dominate. It is important to note that the constants and logarithmic factors, which are encapsulated within the $\tilde{\mathcal{O}}$ notation, are independent of Δ. The proof follows a straightforward approach. When $\Delta = \Omega\left(\sqrt{\frac{\log(n)}{n^{1-\alpha}}}\right)$, the $n^{1-\alpha}\Delta$ term in Theorem 3 takes precedence, and it can be naturally bounded by $\tilde{\mathcal{O}}(n^{1-\alpha})$. On the other hand, when $\Delta = \mathcal{O}\left(\sqrt{\frac{\log(n)}{n^{1-\alpha}}}\right)$, the relevant term is $\mathcal{O}\left(\frac{\log(n)}{\Delta} \wedge n\Delta\right)$, which reaches its maximum at $\Delta = \Theta\left(\sqrt{\frac{\log(n)}{n}}\right)$. Consequently, it can be uniformly bounded by $\tilde{\mathcal{O}}(\sqrt{n})$. Therefore, irrespective of the encountered instance, the regret can always be controlled within the range of $\tilde{\mathcal{O}}(\sqrt{n} \vee n^{1-\alpha})$.

In terms of the inference aspect, Theorem 4 itself holds for any $\Delta = \mathcal{O}(1)$, allowing for direct application in this context. Consequently, the estimation error and the length of the confidence interval can always be controlled within $\tilde{\mathcal{O}}\left(1/\sqrt{n^{1-\alpha}}\right)$. Combining this result with the aforementioned regret outcome, it becomes evident that there is almost no incentive for any decision maker to choose $\alpha > 1/2$. This is because selecting an α larger than $1/2$ would increase the estimation error without further reducing the regret rate. Thus, any EXP3E policy with $\alpha > 1/2$ would be Pareto dominated by EXP3E with $\alpha = 1/2$. This observation highlights the fundamental distinction between $\mathcal{E}_0$ and $\tilde{\mathcal{E}}_0$. Therefore, EXP3E satisfies the sufficient condition, and we can conclusively state that EXP3E achieves Pareto optimality without requiring the assumption $\Delta = \Theta(1)$.

Regrettably, we are unable to establish the necessity of the condition. In Section 2, we can address this by showing that any policy failing to meet the condition will be Pareto dominated by EXP3E with a delicate choice of α. The large-gap assumption enables any $\alpha \in [0,1]$ to yield a Pareto optimal solution. Consequently, EXP3E can cover every point of the curve $e_\nu(n,\hat{\Delta})\sqrt{\mathcal{R}_\nu(n,\pi)} = \Theta(1)$ by varying the value of α. In the absence of the large-gap assumption, as discussed earlier, it becomes meaningful to only choose $\alpha \in [0,1/2]$ for EXP3E. This limitation implies that we cannot exclude the possibility that some Pareto optimal designs may achieve a small regret but incur a significant estimation error on certain instances, thereby violating the specified condition.

7. Discussion

7.1. Alternatives to EXP3E/EXP3EG

In the previous sections, we have demonstrated the Pareto optimality achieved by EXP3E/EXP3EG when considering both online decision-making and ATE inference. This leads to a natural question. Can other efficient MAB algorithms, like UCB or TS, also be modified to achieve Pareto optimality? Furthermore, is there a general framework that can be applied to any efficient MAB algorithm to make it Pareto optimal? In the fully stochastic case explored in Sections 2 and 3, the answer

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is yes. In this section, we will present a straightforward framework with its statistical properties. We will then discuss why we diverge from this framework and choose the current EXP3E/EXP3EG approach.

Based on the insights from the previous sections, it becomes evident that the crucial aspect of designing a Pareto optimal design is to ensure that an efficient MAB algorithm explores all arms, including the suboptimal ones, to gather valuable information. A simple and straightforward approach to achieve this is the explore-then-learn (ETL) strategy. In essence, similar to the well-known explore-then-commit idea in MAB, the ETL strategy consists of two phases.

- Phase 1 (explore). Play each arm uniformly $\Theta(n^{1-\alpha})$ times.
- Phase 2 (learn). Run a regret-optimal MAB algorithm (e.g., UCB, TS, and EXP3).

During phase 2, the regret can be bounded by $\mathcal{O}(\log(n))$ under the large-gap assumption and $\tilde{\mathcal{O}}(\sqrt{n})$ without the large-gap assumption in the worst-case scenarios. In the first phase, the regret is directly bounded by $\mathcal{O}(n^{1-\alpha})$. As for the ATE estimator, a simple sample average of the data collected in the first phase can be employed. By applying standard concentration inequalities, an estimation error of $\mathcal{O}\big(1/\sqrt{n^{1-\alpha}}\big)$ can be guaranteed. Hence, the ETL strategy achieves performance on par with EXP3E and EXP3EG for both the instance classes $\mathcal{E}_0$ and $\tilde{\mathcal{E}}_0$. We can then have the follows.

Corollary 6. *For any given $\alpha \in [0,1]$, under both $\mathcal{E}_0$ and $\tilde{\mathcal{E}}_0$, ETL is Pareto optimal.*

The ETL framework offers clear advantages. It is adaptable to incorporate any well-established efficient MAB policy and can readily leverage future advancements in MAB problems. However, there are concerns that initially deterred us from adopting ETL. First, although ETL is Pareto optimal under the fully stochastic case (i.e., under $\mathcal{E}_0$ and $\tilde{\mathcal{E}}_0$), UCB/TS algorithms face challenges in handling the adversarial reward structure discussed in Section 4. The robustness against adversaries is one of the appealing characteristics of EXP3E/EXP3EG. Second, in many applications, experimenters aim to identify the best treatment as quickly as possible, making uniform exploration at the beginning of the experiment less ideal. One potential solution is to adopt a "learn-then-explore" approach similar to EXP3E and EXP3EG. However, designing the elimination rule, particularly for TS, can be nontrivial. EXP3E and EXP3EG excel in seamlessly navigating all aspects of this work. Even in the batch setting discussed in Section 5, where additional batch design is required, the elimination rule of BaSED closely resembles that of EXP3E and EXP3EG.

There are possible alternative approaches to incorporate exploration of suboptimal arms into standard MAB algorithms. For instance, in the work by Caria et al. (2024), an experiment is designed where each experimental unit is assigned according to the recommendation of TS with a probability of 0.8, and with a probability of 0.2, assignments are made uniformly at random among all treatments. Although Caria et al. (2024) demonstrate strong empirical performance of such a design, they do not provide a theoretical analysis. Nevertheless, this approach presents another viable method for experiment design based on well-known MAB algorithms.

7.2. Alternative to the Error

In previous sections, we primarily focused on the expected error metric, specifically the expected difference between the true treatment effect and the estimated treatment effect. However, in the context of causal inference, the length of confidence intervals at a given confidence level is another important metric of interest. We have addressed the construction of always-valid confidence intervals in Theorem 4. A remaining question is whether the other results, including lower bounds and the sufficient and necessary conditions, still hold when we consider the length of confidence intervals as the second objective of Equation (1). Formally, for all instances ν, given a fixed confidence level δ (say, 0.95), define $l(n,\hat{\Delta},\delta)$ to be the length of the confidence interval (i.e., $\mathbb{P}\big(\Delta_\nu \in \big[\hat{\Delta} - (l(n,\delta)/2), \hat{\Delta} + (l(n,\delta)/2)\big]\big) \geq \delta$). Then, the optimal design of the MAB experiment can be transformed into the following optimization problem with $K = 2$:

$$\min_{(\pi,\hat{\Delta})} \max_{\nu\in\mathcal{E}_0} (\mathcal{R}_\nu(n,\pi), l(n,\delta)). \tag{13}$$

Please note that although we switch our gear to another statistical power metric, the fundamental trade-off between the two objectives still exists. Moreover, the lower bound established in Lemma 1 is still informative here. Equation (4) can be interpreted as follows; for any estimator, there exists a lower bound on the probability of the interval $[\hat{\Delta} - \phi(n), \hat{\Delta} + \phi(n)]$ failing to cover the true value Δ_ν. This implies that regardless of the estimator used, when the length of the interval $2\phi(n)$ is small, the probability of accurately covering Δ_ν inevitably becomes small. Building upon these insights, we can extend the lower bound presented in Theorem 1 as follows.

Theorem 13. *For any admissible pair $(\pi,\hat{\Delta}_n)$ and $\delta > 1/2$, there always exists a hard instance $\nu \in \mathcal{E}_0$ that $l_\nu(n, \hat{\Delta}_n, \delta)\sqrt{\mathcal{R}_\nu(n,\pi)}$ is no less than a constant order: that is,*

$$\inf_{(\pi,\hat{\Delta}_n)} \max_{\nu\in\mathcal{E}_0} \big[l(n,\delta)\sqrt{\mathcal{R}_\nu(n,\pi)}\big] = \Omega(1).$$

Satisfying $\delta > 1/2$ is typically achievable in practice, as common practices often require relatively large values, such as 0.95. In the case of EXP3E, it guarantees a length of confidence interval of $\mathcal{O}\big(1/\sqrt{n^{1-\alpha}}\big)$, which matches the aforementioned lower bound. Moreover, we can readily extend the sufficient and necessary conditions to this scenario, yielding the following.

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Theorem 14. *The admissible pair* $(\pi^*, \hat{\Delta}^*)$ *is Pareto optimal to the minimax multi-objective optimization Problem* (13) *if and only if* $\max_{\nu\in\mathcal{E}_0}[l(n,\delta)\sqrt{\mathcal{R}_\nu(n,\pi^*)}] = \tilde{\mathcal{O}}(1)$.

In summary, the results and intuitions remain almost identical when replacing the error with the length of the confidence interval.

7.3. Neglect of Constants and Logarithm Terms

The final point we want to discuss is the neglect of constants and logarithm terms in defining Pareto optimality. Although this is a common practice in MAB, ignoring these terms can be significant, especially in cases where sample collection is costly. Admittedly, such a simplification is another limitation of our work. It is very challenging to get optimal rates on the dependence of both constant and n. Even for the traditional MAB problem without the added complexity of inference, there is a lack of research addressing the best achievable dependence on constants.

8. Concluding Remarks

In this paper, we statistically investigate the trade-off between the regret and the estimation error of ATE in MAB experiments. We introduce the minimax multi-objective optimization framework and Pareto optimality to formally describe and theoretically analyze such a trade-off. We then derive a sufficient and necessary condition for the Pareto optimal solutions (i.e., $(\max_{i<j\le K} e_\nu(n, \hat{\Delta}^{*(i,j)}))\sqrt{\mathcal{R}_\nu(n,\pi^*)} = \tilde{\mathcal{O}}(1)$ for any instance ν). Moreover, we derive an efficient Pareto optimal design where we can guarantee that $max_{i<j\le K} e_\nu(n, \hat{\Delta}^{(i,j)}) = \mathcal{O}(n^{-(1-\alpha)/2})$ and $\mathcal{R}_\nu(n,\pi) = \mathcal{O}(n^{1-\alpha})$ for any given $\alpha \in [0,1]$ controlling the desired level of trade-off between the two objectives. We also extend our analysis and design to the setting where the outcome of each arm consists of a stochastic treatment effect and an adversarial baseline reward, which shows the robustness of our design. Finally, we examine the batched MAB experimental design where the experimental units must be split into small batches and propose a Pareto optimal design.

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Endnote

[1] Formally, for two positive function $f(n)$ and $g(n)$, we say $f(n) \ge g(n)$ (or $f(n) < g(n)$) if $g(n) = \tilde{O}(f(n))$ (or $f(n) = o(g(n))$). $f(n) = g(n)$ is used to represent $f(n) = \tilde{O}(g(n))$.

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